

Contents

1 Home

2 Cybersecurity First Steps

2.1 Course Overview

2.2 Module 1: Getting Started

2.2.1 Guide 1: Learning Path

2.2.2 Guide 2: What Is Cybersecurity?

2.2.3 Guide 3: Safety and Permission

2.3 Module 2: Getting Started with Kali Linux

2.3.1 Guide 4: What Is Kali?

2.3.2 Guide 5: Download Kali

2.3.3 Guide 6: Create a Live USB

2.3.4 Guide 7: Starting Kali for the First Time

2.3.5 Guide 8: Keeping Kali Up to Date

2.3.6 Guide 9: Shutting Down Kali Safely

2.4 Module 3: Learning Your Way Around Linux

2.4.1 Welcome

2.4.2 Guide 10: Terminal Basics

2.4.3 Guide 11: Files and Folders

2.4.4 Guide 12: Useful Commands

2.4.5 Guide 13: Finding Help

2.4.6 Guide 14: Creating and Managing Files

2.4.7 Guide 15: Understanding Permissions

2.4.8 Module 3 Review

2.4.9 Module 3 Milestone

2.5 Module 4: Understanding Networks

2.5.1 Welcome

2.5.2 Guide 16: What Is a Network?

2.5.3 Guide 17: Wi-Fi vs Ethernet

2.5.4 Guide 18: Understanding IP Addresses

2.5.5 Guide 19: Understanding Ports

2.5.6 Guide 20: How Devices Find Each Other (DNS)

2.5.7 Guide 21: Understanding Your Home Network

2.5.8 Module 4 Review

2.5.9 Module 4 Milestone

2.6 Module 5: Your First Security Tools

2.6.1 Welcome

2.6.2 Guide 22: Why Kali Includes So Many Tools

2.6.3 Guide 23: Meeting Nmap

2.6.4 Guide 24: Meeting Wireshark

2.6.5 Guide 25: Safe Network Discovery

2.6.6 Guide 26: Your First Home Network Inventory

2.6.7 Module 5 Review

2.6.8 Module 5 Milestone

2.7 Resources

2.7.1 Beginner Progress Checklist

2.7.2 Glossary

3 About

4 Teaching Philosophy

5 Learning Framework

6 Downloads

7 Updates

8 Changelog

9 Content Roadmap

10 Style Guide

1 Horizon Learning



Preview Notice

Horizon Learning is currently in early preview. Content is actively being written, reviewed, and improved. Some lessons may change as we receive feedback from early learners.

Learn technology one step at a time.

Horizon Learning is built for people who want to understand technology without getting overwhelmed.

The goal is to make complex topics easier to learn through clear explanations, practical examples, and step-by-step learning paths.

1.1 Current Learning Paths

Cybersecurity First Steps

A beginner-friendly introduction to cybersecurity, Linux, Kali Linux, networking basics, and safe home lab practice.

This path is designed for learners who are curious about cybersecurity but may not have used Linux or command-line tools before.

[Start Cybersecurity First Steps](#)

1.2 Why Horizon Learning?

There is already a lot of technical information online.

The problem is not a lack of information.

The problem is that beginners often do not know:

- Where to start
- What order to learn things in
- Which terms matter
- What commands actually do
- Whether an error is normal
- What to practice next

Horizon Learning is designed to smooth out that process.

1.3 How lessons are written

Guides are designed to:

- Explain ideas in plain English
- Avoid assuming prior experience
- Introduce one concept at a time
- Explain why each step matters
- Include checkpoints so learners know when they are ready to move on
- Encourage safe, responsible practice

1.4 Planned Learning Paths

Future learning paths may include:

- Linux Fundamentals
- Networking Fundamentals
- Git and GitHub
- Python Fundamentals
- Docker Basics
- Self-Hosting Basics
- Home Lab Builder

1.5 Start learning

If you are new here, start with:

[Cybersecurity First Steps](#)

2 Cybersecurity First Steps

2.1 Cybersecurity First Steps



Preview Notice

Cybersecurity First Steps is currently in early preview. This learning path is usable, but not complete. Feedback from early learners will help improve explanations, examples, labs, and downloadable resources.

Cybersecurity First Steps is a beginner-friendly learning path for people who are curious about cybersecurity but are new to Linux, Kali Linux, networking, and command-line tools.

The goal is not to rush into hacking.

The goal is to build understanding step by step.

Who this learning path is for

This path is for beginners who:

- Have little or no Linux experience
- Want to understand Kali Linux
- Want to learn basic cybersecurity tools
- Want safe hands-on practice
- Prefer plain-English explanations
- Want to understand what commands and tools actually do

What you will learn

By the end of this learning path, you should understand:

- What cybersecurity is
- Why permission and safety matter
- What Kali Linux is used for
- How to create a Kali live USB
- Basic Linux terminal commands
- Basic networking concepts
- What Nmap does
- What Wireshark does
- How to perform a safe home Wi-Fi security review

- How to continue learning through legal practice labs

Learning path modules

Module 1: Getting Started

Learn what cybersecurity is, how this path is organized, and why permission matters.

Module 2: Kali Linux

Learn what Kali is, how to download it, how to create a live USB, and how to boot into it.

Module 3: Linux Basics

Learn basic terminal usage, file navigation, and beginner commands.

Module 4: Networking Basics

Learn about networks, Wi-Fi, IP addresses, ports, and services.

Module 5: Beginner Tools

Learn what Kali tools are, then start with Nmap and Wireshark.

Module 6: Safe Practice Labs

Apply what you learned through safe, authorized beginner labs.

Important rule

Only test systems, devices, accounts, and networks that you personally own or have clear permission to test.

If you are unsure whether you have permission, stop.

Recommended pace

Take your time.

A good beginner pace is one or two lessons at a time.

Do not worry about memorizing everything immediately. Focus on understanding the main idea of each lesson before moving on.

Start here

Begin with:

[Guide 1: Learning Path](#)

2.2 Module 1: Getting Started

2.2.1 Learning Path

Goal

Understand the order you should follow as a beginner.

Plain-English explanation

Cybersecurity can feel overwhelming because there are many topics, tools, and opinions.

This guide is designed to avoid that problem.

Instead of jumping straight into advanced tools, you will build the basics first.

Beginner path

Follow this order:

1. Safety and permission
2. What cybersecurity is
3. What Kali Linux is
4. How to download Kali
5. How to create a live USB
6. How to boot into Kali
7. Basic Linux commands
8. Basic networking
9. Kali tool overview
10. Home Wi-Fi security check
11. First Nmap scan
12. Beginner lab roadmap

Why this order matters

Kali Linux includes many advanced tools.

If you open Kali without understanding Linux or networking basics, it can feel confusing fast.

The goal is to learn what the tools are doing, not just copy commands.

Recommended pace

Take your time.

A good pace is one or two lessons at a time.

If a lesson feels confusing, repeat it before moving forward.

Stop and check

You are ready to continue when:

- ☐ You understand the safety rules
- ☐ You know this guide is for authorized learning only
- ☐ You are ready to learn step by step

2.2.2 What Is Cybersecurity?

Goal

Understand what cybersecurity means in plain English.

Plain-English explanation

Cybersecurity is the practice of protecting computers, phones, networks, websites, accounts, and data from misuse or attack.

It includes learning how systems work, how they fail, and how to make them safer.

Cybersecurity is not just hacking

Cybersecurity includes many areas:

- Protecting accounts
- Securing Wi-Fi
- Finding weak passwords
- Updating software
- Detecting suspicious activity
- Investigating incidents
- Testing systems with permission
- Teaching users how to avoid scams

Red team and blue team

You may hear people talk about red team and blue team.

Red team means offensive security. These people test systems by thinking like attackers.

Blue team means defensive security. These people protect systems, monitor alerts, and respond to problems.

Both sides are important.

Beginner focus

As a beginner, your first focus should be:

- Basic computer safety
- Linux basics
- Networking basics
- Safe tool usage
- Legal practice labs

Stop and check

You are ready to continue when:

- ☐ You know cybersecurity is bigger than hacking
- ☐ You understand that permission matters
- ☐ You know both offense and defense exist

2.2.3 Safety and Permission

Goal

Understand where cybersecurity learning is allowed and where it can get you into trouble.

Plain-English explanation

Cybersecurity tools are powerful.

The same tool that helps you check your own network can also be misused against someone else's network.

This guide is only for legal, authorized learning.

Cybersecurity is not about attacking people. It is about understanding systems, finding weaknesses, and learning how to fix or prevent problems.

The main rule

Only test something if:

- You own it
- You control it
- Or you have clear permission from the owner

If the answer is not clearly yes, stop.

Safe places to practice

You can practice on:

- Your own computer
- Your own Wi-Fi network
- Your own router
- A spare router used only for practice
- Local virtual machines
- Training labs made for cybersecurity practice

Do not test

Do not scan, attack, or probe:

- Someone else's Wi-Fi
- School networks
- Work networks
- Public Wi-Fi
- Random websites
- Friends' devices without clear permission
- Neighbor networks
- Any system you do not own or control

Why this matters

Some cybersecurity tools send traffic that can look suspicious or hostile.

Even if your intent is harmless, using those tools on systems you do not own can cause serious problems.

Safe beginner mindset

A good beginner question is not:

“Can I hack this?”

A better question is:

“Do I understand how this works, and am I allowed to test it?”

Stop and check

Before using any tool, ask:

- [] Do I own this system or network?
- [] Do I have permission?
- [] Do I understand what the tool is about to do?
- [] Am I prepared to stop if something looks wrong?

If any answer is no, stop.

What to do next

Continue to the learning path.

2.3 Module 2: Getting Started with Kali Linux

2.3.1 What Is Kali Linux?

Goal

Understand what Kali Linux is and what it is used for.

Why This Matters

Kali Linux can look intimidating at first because it includes many security tools.

This lesson explains what Kali is, what it is not, and why we are starting with a Live USB instead of immediately installing it.

The goal is to understand Kali as a learning environment, not treat it like a magic hacking system.

Your Kali Journey

There are several ways to use Kali Linux.

In this learning path, we will introduce them one at a time so you can choose what works best for you.

We will begin with the safest and easiest option:

- Live USB

Later, we will also cover:

- Live USB with Persistence
- Virtual Machines
- Dual Boot
- Dedicated Installation

For now, you do not need to choose a permanent setup.

Start with the Live USB path, learn the basics, and make a long-term decision later.

Plain-English explanation

Kali Linux is a special version of Linux made for cybersecurity work.

It comes with many tools that can be used for testing networks, analyzing traffic, checking systems, and learning security concepts.

What Kali is good for

Kali can be useful for:

- Learning cybersecurity tools
- Network testing
- Web application testing
- Password auditing in legal lab environments
- Digital forensics practice
- Wireless security learning
- Capture The Flag practice

What Kali is not

Kali is not magic.

It does not automatically make someone a hacker.

It is also not the best everyday operating system for most beginners.

Think of Kali like a toolbox. The tools are useful, but you still need to learn what they do and when to use them.

Live USB

A live USB lets you boot Kali from a USB drive without installing it on the computer.

This is a good beginner option because it lets you explore Kali without replacing Windows.

Stop and check

You are ready to continue when:

- ☐ You know Kali is a cybersecurity-focused Linux system
- ☐ You understand it is a toolbox, not magic
- ☐ You know a live USB can let you try Kali without installing it

2.3.2 Download Kali

Goal

Download Kali Linux safely from the official source.

Why this matters

Kali Linux should only be downloaded from the official Kali website.

Downloading operating system files from random websites is risky because the file could be outdated, modified, or unsafe.

Plain-English explanation

Kali is downloaded as an ISO file.

An ISO file is a disk image. Think of it like a packaged copy of an operating system that can be written to a USB drive.

In the next lesson, that ISO file will be used to create a live USB.

New terms

- **ISO:** A disk image file used to create bootable media.
- **Live USB:** A USB drive that can start an operating system without installing it permanently.
- **Official source:** The real website maintained by the project or company that makes the software.
- **Download mirror:** A server that hosts a copy of a file. Mirrors can be legitimate, but beginners should start from the official download page.

What you need

- A computer with internet access
- A USB drive for the next lesson
- Enough storage space for the ISO file
- Time for the download to finish

Step 1: Go to the official Kali website

Open a web browser.

Search for:

Kali Linux official download



Make sure the result goes to the official Kali website:

kali.org



Do not use random download sites.

Step 2: Open the Downloads page

On the Kali website, open the Downloads section.

You may see several options.

For this beginner path, the goal is to create a live USB, not install Kali permanently.

Look for the live image option.

Step 3: Choose the live image

Choose the live image for your computer type.

Most modern Windows laptops use 64-bit Intel or AMD processors.

That is usually listed as:

64-bit



or:

amd64



The term `amd64` does not mean it only works on AMD computers. It is commonly used to mean 64-bit Intel/AMD compatible systems.

Step 4: Download the ISO

Download the ISO file.

Save it somewhere easy to find, such as:

Downloads



Do not rename the file yet.

Step 5: Wait for the download to finish

The file may be large.

Let the download finish completely before moving to the next lesson.

If the download fails, delete the partial file and download it again.

Step 6: Optional verification

More advanced users can verify the downloaded file using checksums or signatures.

Verification helps confirm that the downloaded file matches what Kali published.

For the first version of this beginner guide, the safest simple rule is:

- Download only from the official Kali website
- Do not use third-party download sites
- Do not use files sent by strangers
- Re-download if something looks wrong

A future version of this guide may include a full beginner-friendly verification walkthrough.

Common mistakes

- Downloading from an unofficial website
- Downloading the installer when the lesson expects the live image
- Deleting the ISO before creating the USB
- Forgetting where the file was saved
- Trying to open the ISO like a normal app

Stop and check

You are ready to continue when:

- ☐ You downloaded Kali from the official Kali website.
- ☐ You downloaded the live image.
- ☐ You know where the ISO file is saved.
- ☐ The download finished completely.

- ☐ You understand the ISO will be written to a USB drive in the next lesson.

What to do next

Continue to Lesson 6: Create a Live USB.

2.3.3 Lesson 6: Create a Live USB

Estimated Time: 30–45 minutes

Difficulty: Beginner

Platform Support: Windows, Linux, macOS

Learning Objectives

By the end of this lesson, you should be able to:

- Explain what a live USB is.
 - Understand why creating a live USB erases the USB drive.
 - Choose the correct instructions for your operating system.
 - Create a Kali live USB using a beginner-friendly tool.
 - Avoid the most common live USB mistakes.
-

Goal

Create a USB drive that can boot into Kali Linux without installing Kali permanently on your computer.

Why This Matters

A live USB lets you try Kali without replacing your current operating system.

This is a good beginner option because you can explore Kali, use the terminal, and follow beginner labs without committing to a full installation.



Warning

Creating a live USB will erase the selected USB drive.

Do not use a USB drive that contains important files.

Plain-English Explanation

Normally, your computer starts from its internal drive.

That internal drive usually starts Windows, Linux, or macOS.

A live USB lets your computer start from the USB drive instead.

The USB drive contains Kali Linux, so the computer can temporarily run Kali from the USB.

This does **not** mean Kali is installed permanently.

New Terms

- **Live USB:** A USB drive that can start an operating system.
 - **Boot:** Start the computer.
 - **Bootable:** Able to start a computer.
 - **ISO:** A disk image file used to create bootable media.
 - **Flash:** Write an operating system image to a USB drive.
 - **USB writing tool:** A program that writes an ISO file to a USB drive correctly.
-

Before You Start

You need:

- The Kali ISO file from Lesson 5.
- A USB drive you are willing to erase.
- A computer running Windows, Linux, or macOS.
- A USB writing tool.



If you have multiple USB drives plugged in, remove the ones you are not using.

This reduces the chance of selecting the wrong drive.

Choose Your Platform

This task can be completed on Windows, Linux, or macOS.

Choose the section that matches the computer you are currently using.

2.3.3 Windows

Recommended beginner tools

On Windows, beginner-friendly options include:

- Balena Etcher
- Rufus

Kali's official Windows USB documentation includes both Etcher and Rufus options. [:contentReference\[oaicite:1\]{index=1}](#)

For beginners, Balena Etcher is often simpler because it has fewer choices.

Rufus is also popular and powerful, but it shows more options.

Windows Steps

1. Insert the USB drive.
2. Open Balena Etcher or Rufus.
3. Select the Kali ISO file.
4. Select the USB drive.
5. Carefully confirm the selected USB drive.
6. Start the write or flash process.
7. Confirm the warning that the USB drive will be erased.
8. Wait until the tool says the process is complete.
9. Safely eject the USB drive.



Warning

The most important step is selecting the correct USB drive.

If you choose the wrong drive, you can erase the wrong device.

2.3.3 Linux

Recommended beginner tools

Linux users may have several options depending on their distribution.

Common graphical tools include:

- Balena Etcher
- Popsicle
- GNOME Disks
- KDE ISO Image Writer
- Linux Mint USB Image Writer

Kali's official Linux USB documentation includes a terminal-based `dd` method, but that method can erase the wrong drive if used incorrectly. [:contentReference\[oaicite:2\]{index=2}](#)

For beginners, a graphical USB writing tool is usually safer.

Linux Mint

Linux Mint often includes a tool called:

USB Image Writer



Basic steps:

1. Insert the USB drive.
2. Open the application menu.
3. Search for:

USB Image Writer



4. Open USB Image Writer.
5. Select the Kali ISO file.
6. Select the USB drive.
7. Confirm the selected drive is correct.
8. Start writing the image.
9. Wait until the process finishes.

10. Safely eject the USB drive.

Pop!_OS

Pop!_OS may use a tool called:

Popsicle



Basic steps:

1. Insert the USB drive.
2. Open the application menu.
3. Search for:

Popsicle



4. Open Popsicle.
 5. Select the Kali ISO file.
 6. Select the USB drive.
 7. Confirm the selected drive is correct.
 8. Start the write process.
 9. Wait until it finishes.
 10. Safely eject the USB drive.
-

GNOME-based Linux distributions

Some Linux systems include:

GNOME Disks



Basic steps:

1. Insert the USB drive.
2. Open Disks.
3. Select the USB drive from the left side.
4. Open the menu.
5. Choose an option like:

6. Select the Kali ISO file.
 7. Confirm the USB drive is correct.
 8. Start the restore/write process.
 9. Wait until it finishes.
 10. Safely eject the USB drive.
-

KDE-based Linux distributions

Some KDE systems may include:

Basic steps:

1. Insert the USB drive.
 2. Open KDE ISO Image Writer.
 3. Select the Kali ISO file.
 4. Select the USB drive.
 5. Confirm the USB drive is correct.
 6. Start writing.
 7. Wait until complete.
 8. Safely eject the USB drive.
-

Advanced Linux option: dd

The `dd` command can create a live USB, but it is dangerous for beginners because it can overwrite the wrong disk.

This guide does **not** recommend `dd` for first-time learners.



Warning

Do not use `dd` unless you understand Linux disk names and are certain which device is your USB drive.

Choosing the wrong device can erase important data.

2.3.3 macOS

Recommended beginner tool

For macOS beginners, the simplest option is usually:

- Balena Etcher

Kali's official macOS USB documentation includes both Etcher and terminal-based methods.

[:contentReference\[oaicite:3\]{index=3}](#)

For beginners, use Etcher first.

macOS Steps with Balena Etcher

1. Insert the USB drive.
 2. Open Balena Etcher.
 3. Select the Kali ISO file.
 4. Select the USB drive.
 5. Carefully confirm the selected USB drive.
 6. Start the flash process.
 7. Enter your macOS password if prompted.
 8. Wait until the process finishes.
 9. Eject the USB drive safely.
-

Advanced macOS option

macOS can also create bootable USB drives from the terminal.

This is an advanced method.

It is not recommended for first-time learners because selecting the wrong disk can erase data.



Warning

Terminal-based disk writing on macOS can erase the wrong drive if used incorrectly.

Use a graphical tool first unless you are comfortable identifying disks safely.

2.3.3 After Writing the USB

When the process finishes, the USB drive may look strange to your operating system.

Windows, Linux, or macOS may show warnings, extra partitions, or unreadable sections.

This can be normal.

The USB is now designed to boot Kali, not act like a normal storage drive.

2.3.3 Common Mistakes

- Selecting the wrong USB drive.
 - Using a USB drive that contains important files.
 - Copying the ISO file onto the USB instead of writing/flashing it.
 - Removing the USB before the process completes.
 - Panicking when the USB looks strange afterward.
 - Choosing advanced terminal methods too early.
-

2.3.3 Troubleshooting

The USB writing tool does not see the USB drive

Try:

- Reinsert the USB drive.
- Try another USB port.
- Restart the USB writing tool.
- Try another USB drive.

The process fails

Try:

- Re-download the Kali ISO.
- Try a different USB drive.

- Use another USB writing tool.
- Make sure no other program is using the USB drive.

The USB looks unreadable afterward

This may be normal.

A bootable Linux USB may not look like a regular storage drive.

2.3.3 Try It Yourself

Create a Kali live USB using the section for your operating system.

Write down:

- Operating system used:
 - Tool used:
 - USB drive size:
 - Did the process finish successfully?
 - Any error messages:
-

2.3.3 Reflection Questions

Try answering these without looking back:

1. Why does creating a live USB erase the USB drive?
 2. Why is selecting the correct USB drive important?
 3. Why is copying the ISO file normally not enough?
 4. Why are graphical tools recommended for beginners?
 5. What should you do if the USB looks strange afterward?
-

2.3.3 Stop and Check

You are ready to continue when:

- [] You understand the USB drive was erased.

- ☐ You selected the correct USB drive.
 - ☐ The writing tool completed successfully.
 - ☐ You safely removed or ejected the USB drive.
 - ☐ You are ready to boot from the USB.
-

2.3.3 Lesson Summary

In this lesson you learned:

- A live USB lets you run Kali without permanently installing it.
 - Creating a live USB erases the selected USB drive.
 - Windows, Linux, and macOS can all create live USB drives.
 - Beginners should prefer graphical USB writing tools.
 - The USB may look unusual afterward, and that can be normal.
-

2.3.3 What To Do Next

Continue to **Lesson 7 – First Boot**.

In the next lesson, you will learn how to start your computer from the Kali USB.

2.3.4 Lesson 7: Starting Kali for the First Time

Estimated Time: 25–35 minutes

Difficulty: Beginner

Platform Support: Windows, Linux, and macOS computers using a Kali Live USB

Learning Objectives

By the end of this lesson you should be able to:

- Explain what "booting" means.
 - Understand the difference between a Boot Menu and Boot Order.
 - Recognize BIOS and UEFI.
 - Start your computer from the Kali Live USB.
 - Choose the correct boot option.
 - Become comfortable with the Kali desktop.
-

Goal

Start Kali Linux from the Live USB without installing it on your computer.

Why This Matters

This is the moment many new learners are nervous about.

That is completely normal.

The good news is that you are **not** replacing your current operating system.

Today you are simply asking your computer to start from the USB drive instead of its internal drive.

When you remove the USB and restart normally, your computer should return to its usual operating system.

✓ **Success**

Congratulations!

You have already completed the hardest preparation steps.

Now it's time to see Kali Linux for the first time.

2.3.4 Plain-English Explanation

When you press the power button on a computer, it doesn't immediately start Windows, Linux, or macOS.

Instead, the computer first checks its hardware and decides **where to start**.

Normally, it starts from the internal storage drive.

Today, we're going to tell it to start from the Kali USB instead.

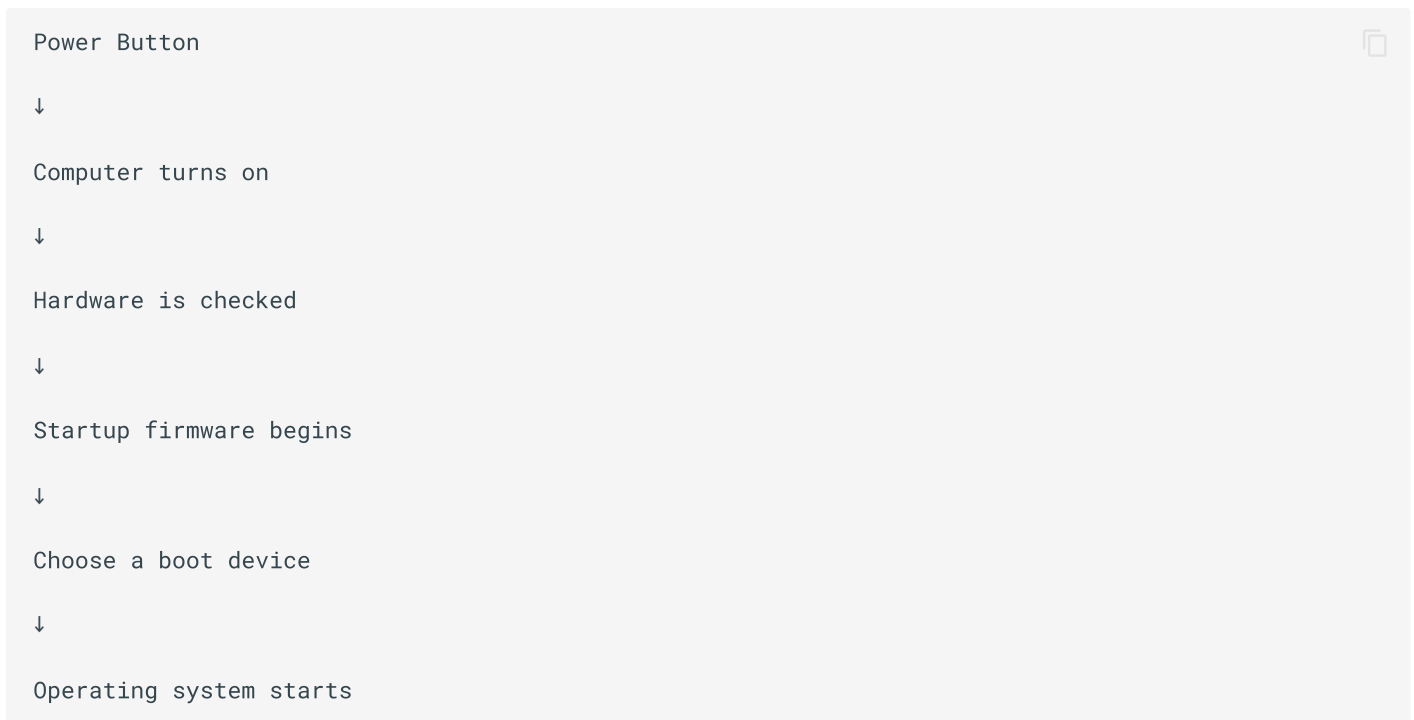
That's all "booting from a USB" really means.

2.3.4 New Terms

- Boot
- Bootable USB
- BIOS
- UEFI
- Boot Menu
- Boot Order
- Live System

2.3.4 What Happens When a Computer Starts?

A simplified version looks like this:



Normally, your computer chooses the internal drive.

Today, you'll temporarily choose the Kali USB.

2.3.4 BIOS vs UEFI

You may hear both of these terms.

BIOS

Older computers commonly used BIOS.

UEFI

Most modern computers use UEFI.

For this lesson, you only need to know one thing:

Both help your computer start.

You do **not** need to become an expert on firmware today.



Tip

Don't worry if you're unsure whether your computer uses BIOS or UEFI.

The learning process is nearly identical.

2.3.4 Boot Menu vs Boot Order

These are often confused.

Boot Menu	Boot Order
Temporary choice	Permanent preference
Used once	Used every startup
Recommended for beginners	Usually leave unchanged

For this course...

Use the Boot Menu whenever possible.

It lets you choose the USB for this startup only.

Why We Are Choosing Live System

The Kali boot menu may show install options.

For this learning path, choose:

Live System



We are choosing Live System because it lets you explore Kali without making permanent changes to your computer.

Do not choose Installer or Graphical Installer for this course unless a later lesson specifically tells you to.

We will compare long-term setup options later, including persistent USB, virtual machines, dual boot, and full installation.

2.3.4 Common Boot Menu Keys

Different manufacturers use different keys.

Some common examples include:

- F12
- F11
- F10
- F9
- Esc
- Del

 Note

If you're unsure which key your computer uses, check the manufacturer's support page or user manual.

It is completely normal to look this up.

2.3.4 Secure Boot

Many computers include a feature called Secure Boot.

Secure Boot helps protect the startup process by allowing only trusted boot software to run.

Some computers can boot Kali without any changes.

Others may require additional configuration depending on the hardware and firmware.

For this lesson:

Do not change Secure Boot settings unless you actually encounter a problem.

 Warning

Avoid disabling security features simply because a tutorial says to.

Understand **why** a change is needed before making it.

2.3.4 Starting the Kali USB

Insert your Kali USB drive.

Restart the computer.

Open the Boot Menu.

Select the USB drive.

After a few moments, you should see the Kali boot menu.

2.3.4 The Kali Boot Menu

You may see options similar to:

- Live System
- Live System (Failsafe)
- Installer
- Graphical Installer

For this learning path:

Choose:

 Live System

This allows you to explore Kali without installing it.

Behind the Curtain

The Live System runs directly from the USB drive.

Nothing is permanently installed on your internal storage.

This makes it an excellent way to learn safely.

2.3.4 Welcome to Kali

After Kali finishes loading, take a moment to look around.

You should see:

- Desktop
- Applications menu

- Terminal
- File manager
- Network icon
- Clock
- Power menu

Today, don't worry about learning everything.

Simply become comfortable looking around.

2.3.4 Your First Goal

Try these simple tasks:

- Move the mouse.
- Open the Applications menu.
- Open the Terminal.
- Close the Terminal.
- Open the File Manager.
- Return to the desktop.

That's enough for today.

2.3.4 Common Problems

The USB Doesn't Appear

Try:

- Reinsert the USB drive.
 - Restart the computer.
 - Use a different USB port.
 - Confirm the USB was written successfully.
-

The Computer Starts Windows Instead

Usually this means the USB wasn't selected.

Restart and try the Boot Menu again.

Black Screen

Wait several minutes.

If nothing changes:

- Restart.
 - Try Live System (Failsafe).
 - Verify the USB creation process.
-

Keyboard Doesn't Work

Try another USB port.

Some very new hardware may require a newer Kali image.

2.3.4 Try It Yourself

Complete the following:

- Boot Kali from the USB.
- Reach the desktop.
- Open the Applications menu.
- Open and close the Terminal.
- Open the File Manager.
- Shut down Kali from the menu.

Do not install Kali.

Simply explore.

2.3.4 Reflection Questions

Without looking back:

1. What does "booting" mean?
 2. Why do we use the Boot Menu instead of changing Boot Order?
 3. What is Secure Boot?
 4. Why are we choosing Live System instead of Installer?
 5. Why is using a Live USB a safe way to learn?
-

2.3.4 Stop and Check

You are ready to continue when:

- ☐ Kali started successfully.
 - ☐ You reached the desktop.
 - ☐ The mouse works.
 - ☐ The keyboard works.
 - ☐ You opened the Applications menu.
 - ☐ You opened the Terminal.
 - ☐ You opened the File Manager.
 - ☐ You shut Kali down normally.
-

2.3.4 Lesson Summary

In this lesson you learned:

- What "booting" means.
 - The difference between Boot Menu and Boot Order.
 - The purpose of BIOS and UEFI.
 - Why we use Live System.
 - How to explore Kali without installing it.
-

2.3.4 What To Do Next

Continue to **Lesson 8 – Updating Kali Linux**.

You'll learn:

- Why updates matter.
- How Linux installs software.
- How package managers work.
- How to safely update Kali.

2.3.5 Lesson 8: Updating Kali Linux

Estimated Time: 20–30 minutes

Difficulty: Beginner

Platform Support: Kali Linux

Learning Objectives

By the end of this lesson, you should be able to:

- Explain why updates matter.
 - Understand what a package manager does.
 - Recognize the difference between updating package lists and installing updates.
 - Safely update Kali Linux.
 - Understand why live USB changes may not be saved.
-

Goal

Learn how Kali Linux updates software and how to safely run basic update commands.

Why This Matters

Kali includes many tools.

Those tools change over time.

Updates can fix bugs, improve security, and add newer versions of software.

Before using Kali seriously, it is important to understand how updates work.

Warning

If you are using Kali in normal Live USB mode without persistence, updates may not be saved after reboot.

That is normal.

Persistence and installed systems will be covered later.

Plain-English Explanation

Most Linux systems use a package manager.

A package manager helps install, update, and remove software.

On Kali, the package manager is based on Debian's package system.

The command you will use most often is:

```
apt
```



Think of `apt` as a software manager for the terminal.

New Terms

- **Package:** A bundle of software files.
- **Repository:** A trusted software source.
- **Package manager:** A tool that installs and updates software.
- **apt:** A command-line package manager used by Kali and Debian-based Linux systems.
- **Update package list:** Refresh the list of available software versions.
- **Upgrade packages:** Install newer versions of software.

Before You Start

Make sure:

- Kali is running.
- You are connected to the internet.
- You can open the Terminal.



Tip

If you are using a Live USB, this is still useful practice even if the updates disappear after reboot.

The goal right now is to understand the process.

Step 1: Open the Terminal

Open the Terminal from the Kali desktop.

You should see a prompt waiting for a command.

Step 2: Update the Package List

Run:

```
sudo apt update
```



This does **not** install updates yet.

It asks Kali to check the configured software repositories and refresh the list of available package versions.

You may be asked for a password.

If you are using the default Kali live environment, the default user is often:

```
kali
```



The password is often:

```
kali
```



Note

Default usernames and passwords can change between versions.

If the default password does not work, check the current official Kali documentation for the version you are using.

Step 3: Read the Output

You may see lines that start with words like:

```
Hit
Get
Ign
Err
```



Beginner meaning:

- **Hit** usually means Kali already has current information from that source.
- **Get** means Kali downloaded updated package information.
- **Err** means something went wrong.
- **Ign** means something was ignored.

Do not panic if the output looks busy.

Linux package managers often show a lot of text.

Step 4: Upgrade Installed Packages

After the package list updates successfully, run:

```
sudo apt upgrade
```



Kali may show a list of packages to upgrade.

It may ask:

```
Do you want to continue? [Y/n]
```



If you are ready to continue, type:

```
y
```



Then press Enter.

Step 5: Wait

Updates may take several minutes.

Do not close the terminal while updates are running.

Do not remove the USB drive while Kali is running.

Step 6: Clean Up Unneeded Packages

Sometimes Linux may tell you that some packages are no longer needed.

You may see a suggestion like:

```
sudo apt autoremove
```



This removes packages that are no longer required.

For now, do not run cleanup commands unless you understand why you are running them.

We will keep this lesson focused on the basic update process.

Important Live USB Note

If you are using Kali in normal live mode, changes usually disappear after reboot.

That means:

- Updates may not remain installed.
- Files may not remain saved.
- Settings may reset.

This is expected.

A normal live USB is meant for temporary use.

Behind the Curtain

A live USB loads the operating system from the USB drive.

Unless persistence is configured, changes are stored temporarily and are lost when the system shuts down.

Common Mistakes

- Thinking `apt update` installs updates.
- Closing the terminal while updates are running.
- Removing the USB drive while Kali is running.
- Assuming updates are saved in normal live mode.

- Running random install commands from the internet.
-

Troubleshooting

No Internet Connection

If `sudo apt update` fails, first check your internet connection.

Try opening a website in the browser.

If Wi-Fi is not connected, connect to Wi-Fi from the network icon.

Permission Error

If a command says permission denied, make sure you used:

```
sudo
```



before the command.

Example:

```
sudo apt update
```



Lock Error

Sometimes you may see a message about a lock file.

This can happen if another package process is already running.

Wait a few minutes and try again.

Do not delete lock files unless you understand what caused the problem.

Try It Yourself

Run these commands in order:


```
sudo apt update
sudo apt upgrade
```



Write down:

- Did `apt update` complete successfully?
 - Did `apt upgrade` find updates?
 - Did Kali ask for confirmation?
 - Did anything confuse you?
-

Reflection Questions

Try answering these without looking back:

1. What does a package manager do?
 2. What is the difference between `apt update` and `apt upgrade` ?
 3. Why do updates matter?
 4. Why might updates disappear after rebooting a live USB?
 5. Why should you be careful copying install commands from the internet?
-

Stop and Check

You are ready to continue when:

- ☐ You opened the Terminal.
 - ☐ You understand what `apt` does.
 - ☐ You understand the difference between `apt update` and `apt upgrade` .
 - ☐ You ran the update commands or understand how they work.
 - ☐ You understand that Live USB changes may not be saved.
-

Lesson Summary

In this lesson you learned:

- Kali uses a package manager to update software.
- `sudo apt update` refreshes package information.
- `sudo apt upgrade` installs available updates.

- Live USB changes may not persist after reboot.
 - Updates are important, but commands should still be understood before running them.
-

What To Do Next

Continue to **Lesson 9 – Shutting Down Kali Safely**.

You will learn how to shut down Kali properly and avoid damaging files or removing the USB too early.

2.3.6 Lesson 9: Shutting Down Kali Safely

Estimated Time: 15–20 minutes

Difficulty: Beginner

Platform Support: Kali Linux

Learning Objectives

By the end of this lesson, you should be able to:

- Explain why proper shutdown matters.
 - Shut down Kali from the desktop menu.
 - Shut down Kali from the terminal.
 - Understand when it is safe to remove the USB drive.
 - Return to your normal operating system safely.
-

Goal

Learn how to shut down Kali properly and safely remove the Live USB.

Why This Matters

A Live USB is still running an operating system.

Removing the USB drive too early can cause problems.

A proper shutdown gives Kali time to close programs, finish writing files, and stop safely.



Warning

Do not remove the Kali USB while Kali is still running.

Shut down first, wait for the computer to power off, and then remove the USB.

Plain-English Explanation

When Kali is running from a USB drive, the USB is acting like the system drive.

That means Kali may be reading from it or writing to it while the system is running.

If you pull the USB out too early, it can interrupt the operating system.

A clean shutdown prevents that.

New Terms

- **Shutdown:** Turn the operating system off safely.
 - **Restart:** Shut down and immediately start again.
 - **Unmount:** Safely disconnect a drive from the operating system.
 - **Live USB:** A USB drive used to start and run an operating system.
 - **Power menu:** The menu used to shut down, restart, or log out.
-

Method 1: Shut Down from the Desktop

This is the recommended beginner method.

1. Look for the power icon or system menu.
2. Open the menu.
3. Choose shutdown, power off, or log out.
4. Confirm shutdown if prompted.
5. Wait for the screen to go dark or the computer to power off.
6. Remove the USB drive after the computer is off.



Tip

Menu labels can vary slightly depending on the Kali desktop version.

Look for words like **Power Off**, **Shutdown**, or **Restart**.

Method 2: Shut Down from the Terminal

You can also shut down from the terminal.

Open Terminal and run:

```
sudo poweroff
```



This asks Kali to shut down safely.

You may also see this command used:

```
sudo shutdown now
```



Both are common Linux shutdown commands.

When to Remove the USB

Remove the USB only after:

- Kali has fully shut down.
- The screen is off or the computer has powered down.
- The system is no longer actively running from the USB.

If the computer restarts instead of shutting down, remove the USB before it starts booting again.

Returning to Your Normal Operating System

To return to your normal operating system:

1. Shut down Kali.
2. Remove the USB drive.
3. Turn the computer back on.

The computer should start normally from its internal drive.

That usually means:

- Windows
- Linux
- macOS

depending on the computer.

Live USB Reminder

If you are using Kali in normal Live USB mode, changes may not be saved.

That means files, updates, and settings may disappear after shutdown.

This is expected.

Note

A normal Live USB is temporary.

Persistence can be added later if you want Kali to remember changes between restarts.

Common Mistakes

- Pulling the USB drive out while Kali is running.
- Holding the power button instead of shutting down normally.
- Forgetting that live mode may not save changes.
- Restarting and accidentally booting back into Kali.
- Thinking Windows or the normal operating system was deleted.

Troubleshooting

The Computer Boots Back into Kali

This usually means the USB was still plugged in.

Shut down again, remove the USB, and start the computer normally.

The Computer Shows a Boot Menu Again

Choose the internal drive.

If unsure, shut down, remove the USB, and turn the computer back on.

Kali Freezes During Shutdown

Wait a few minutes.

If nothing changes, hold the power button as a last resort.

This should not be the normal method, but it may be necessary if the system is completely frozen.

Try It Yourself

Practice shutting down Kali safely.

Use either:

- Desktop power menu
- Terminal command

Then remove the USB and start the computer normally.

Write down:

- Which shutdown method did you use?
 - Did the system power off normally?
 - Did your normal operating system start after removing the USB?
 - Did anything unexpected happen?
-

Reflection Questions

Try answering these without looking back:

1. Why should you shut down before removing the USB?
 2. What does `sudo poweroff` do?
 3. Why might changes disappear after shutting down a Live USB?
 4. How do you return to your normal operating system?
 5. What should you do if the computer boots back into Kali?
-

Stop and Check

You are ready to continue when:

- ☐ You know how to shut down Kali from the desktop.
 - ☐ You know how to shut down Kali from the terminal.
 - ☐ You understand when it is safe to remove the USB.
 - ☐ You understand that Live USB changes may not be saved.
 - ☐ You can return to your normal operating system.
-

Lesson Summary

In this lesson you learned:

- A Live USB should not be removed while Kali is running.
 - Proper shutdown helps prevent problems.
 - Kali can be shut down from the desktop or terminal.
 - Normal Live USB mode may not save changes.
 - Removing the USB lets the computer return to its usual operating system.
-

What To Do Next

Continue to **Lesson 10 – Terminal Basics**.

You have now completed the basic Kali setup module and are ready to start learning how to use Linux.

2.4 Module 3: Learning Your Way Around Linux

2.4.1 Learning Your Way Around Linux

Welcome!

You've successfully started Kali Linux and explored the desktop.

That is a bigger accomplishment than many people realize.

Now it's time to become comfortable using Linux itself.

Don't worry—you are **not** expected to memorize dozens of commands.

Our goal is much simpler:

Understand how Linux works and become confident using it.

What You'll Learn

In this module you will learn how to:

- Use the terminal with confidence.
 - Navigate the Linux file system.
 - Understand where your files are stored.
 - Find help when you need it.
 - Work safely with files and directories.
 - Understand why Linux permissions exist.
-

Why This Matters

Almost every cybersecurity tool assumes you're comfortable using Linux.

The better you understand Linux, the easier every future lesson will become.

Think of this module as learning the language before trying to write a novel.



Progress, Not Perfection

You do **not** need to memorize every command.

Professional Linux users regularly look up commands and documentation.

Your goal is to understand what commands do and know when to use them.

Module Roadmap

In this module you'll complete:

1. Terminal Basics
2. Files and Folders
3. Useful Commands
4. Finding Help
5. Creating and Managing Files
6. Understanding Permissions

Each lesson builds on the previous one.

Take your time.

By the End of This Module

You'll be able to:

- Open the terminal without feeling intimidated.
 - Move around the Linux file system.
 - Find files and directories.
 - Create, copy, move, and remove files safely.
 - Understand why Linux protects certain files.
 - Feel ready to begin using cybersecurity tools.
-

Let's Get Started

Continue to **Lesson 10 – Terminal Basics**.

2.4.2 Lesson 10: Terminal Basics

Estimated Time: 20–30 minutes

Difficulty: Beginner

Platform Support: Kali Linux

Learning Objectives

By the end of this lesson, you should be able to:

- Explain what the terminal is.
 - Open the terminal in Kali.
 - Understand what a command is.
 - Run a few safe beginner commands.
 - Recognize basic command output.
 - Avoid blindly copying commands from the internet.
-

Goal

Become comfortable opening the terminal and running your first basic Linux commands.

Why This Matters

Many cybersecurity tools are controlled from the terminal.

If the terminal feels confusing, Kali will feel confusing.

You do not need to become an expert today. The goal is to make the terminal feel less intimidating.



Tip

The terminal is not magic.

It is just another way to tell the computer what you want it to do.

Plain-English Explanation

The terminal is a text-based way to control the computer.

Instead of clicking buttons, you type commands.

A command is an instruction.

The computer reads the command, does what it was told, and then shows output.

New Terms

- **Terminal:** A window where you type commands.
 - **Command:** An instruction typed into the terminal.
 - **Prompt:** The text shown before where you type.
 - **Output:** The response shown after a command runs.
 - **Directory:** Another word for folder.
 - **Current directory:** The folder you are currently working in.
-

What the Prompt Means

When you open the terminal, you may see something like:

```
kali@kali:~$
```



This is called the prompt.

It usually shows:

- Your username
- The computer name
- Your current location
- A symbol showing the terminal is waiting for a command

You do not need to fully understand every part yet.

For now, just know that the prompt means:

■ The terminal is ready.

Step 1: Open the Terminal

In Kali, open the terminal by:

1. Looking for the Terminal icon.
2. Or opening the Applications menu.
3. Searching for Terminal.
4. Clicking the Terminal app.

A terminal window should open.

Step 2: Run Your First Command

Type:

```
pwd
```



Press Enter.

This command means:

```
print working directory
```



It shows where you are in the file system.

You may see something like:

```
/home/kali
```



That means you are in the Kali user's home folder.

Step 3: List Files and Folders

Type:

```
ls
```



Press Enter.

This command lists files and folders in your current directory.

If nothing appears, the folder may be empty.

That is normal.

Step 4: Clear the Screen

Type:

```
clear
```



Press Enter.

This clears the terminal screen.

It does not delete files.

It only cleans up what you see in the terminal window.

Step 5: Check Your Username

Type:

```
whoami
```



Press Enter.

This shows the current user.

You may see:

```
kali
```



Warning

Do not copy and paste random commands from the internet.

Some commands can delete files, change settings, install unsafe software, or break things.

If you do not understand what a command does, pause and look it up first.

Behind the Curtain

When you press Enter, the shell reads your command and tries to run it.

A shell is the program that accepts typed commands and sends them to the operating system.

You do not need to memorize that today.

For now, remember:

Terminal is the window. Shell is the command interpreter inside it.

Try It Yourself

Run these commands in order:

```
pwd
ls
whoami
clear
pwd
```



Notice that `clear` did not move you or delete anything.

It only cleared the screen.

Reflection Questions

Try answering these without looking back:

1. What is the terminal?
 2. What is a command?
 3. What does `pwd` show?
 4. What does `ls` show?
 5. Why should you avoid copying random commands from the internet?
-

Common Mistakes

- Typing uppercase letters when the command should be lowercase.
- Forgetting to press Enter.
- Thinking `clear` deletes files.
- Copying commands without understanding them.

- Getting nervous when the terminal shows a lot of text.
-

Troubleshooting

Command Not Found

If a command says:

```
command not found
```



Check for spelling mistakes.

Linux commands are usually lowercase.

For example:

```
pwd
```



is not the same as:

```
PWD
```



Nothing Happens

Some commands only show output when something important happens.

If the prompt returns and there is no error, the command may have completed successfully.

Stop and Check

You are ready to continue when:

- ☐ You can open the terminal.
- ☐ You can run `pwd`.
- ☐ You can run `ls`.
- ☐ You can run `whoami`.
- ☐ You understand that `clear` only clears the screen.
- ☐ You understand that commands should not be blindly copied.

Lesson Summary

In this lesson you learned:

- The terminal lets you control Linux by typing commands.
 - A command is an instruction.
 - `pwd` shows your current folder.
 - `ls` lists files and folders.
 - `whoami` shows the current user.
 - `clear` clears the terminal screen.
 - Commands should be understood before they are copied and run.
-

What To Do Next

Continue to **Lesson 11 – Files and Folders**.

Next, you will learn how Linux organizes files and how to move through folders from the terminal.

2.4.3 Lesson 11: Files and Folders

Estimated Time: 25–35 minutes

Difficulty: Beginner

Platform Support: Kali Linux

Learning Objectives

By the end of this lesson, you should be able to:

- Explain how Linux organizes files.
 - Understand the difference between a file and a directory.
 - Navigate between directories.
 - List the contents of directories.
 - Understand absolute and relative paths.
-

Goal

Learn how Linux stores information and how to move around the file system.

Why This Matters

Everything in Linux is organized into directories.

If you know where you are, you'll rarely feel lost.

Many cybersecurity tools expect you to know where files are stored.

Understanding the file system is one of the biggest confidence boosters for new Linux users.

Plain-English Explanation

Think of your computer like a filing cabinet.

The cabinet contains drawers.

The drawers contain folders.

The folders contain documents.

Linux works the same way.

Directories (folders) hold files and other directories.

New Terms

- File
 - Directory
 - Path
 - Root Directory
 - Home Directory
 - Absolute Path
 - Relative Path
-

The Linux File System

At the very top is the root directory.

It is written as:

```
/
```



Everything else starts here.

For example:

```
/
```

```
  home/
```

```
    kali/
```

```
      Documents/
```

```
      Downloads/
```

```
      Pictures/
```



Behind the Curtain

Unlike Windows, Linux does not use drive letters like C:\ or D:.

Everything begins at the single root directory:

```
/
```



Your Home Directory

Most of your work will happen inside:

```
/home/kali
```



This is your personal workspace.

Think of it as your office inside a larger building.

Step 1: Where Am I?

Run:

```
pwd
```



You should see something similar to:

```
/home/kali
```



Step 2: What's Here?

Run:

```
ls
```



You should see files and folders in your current location.

Step 3: Move Into a Folder

Run:

```
cd Documents
```

If the folder exists, you will move into it.

Check your location:

```
pwd
```

Step 4: Go Back

Run:

```
cd ..
```

The two dots mean:

Go up one directory.

Run:

```
pwd
```

You should be back where you started.

Step 5: Return Home

No matter where you are, you can return home with:

```
cd ~
```

The tilde (~) is a shortcut for your home directory.

Absolute vs Relative Paths

An absolute path starts from the root.

Example:

```
/home/kali/Documents
```

A relative path starts from your current location.

Example:

```
Documents
```

Both may reach the same destination.

Common Commands

Command	Purpose
pwd	Show current directory
ls	List files
cd folder	Enter folder
cd ..	Go up one level
cd ~	Return home

Try It Yourself

Practice this sequence:

```
pwd
ls
cd Documents
pwd
cd ..
pwd
cd ~
pwd
```

Try saying what each command does before pressing Enter.

Reflection Questions

1. What is the root directory?
 2. What is your home directory?
 3. What does `cd ..` do?
 4. What is the difference between an absolute path and a relative path?
 5. Why is knowing your current directory important?
-

Common Mistakes

- Forgetting spaces in commands.
 - Trying to enter a folder that does not exist.
 - Confusing files with directories.
 - Thinking `cd ..` deletes something.
-

Troubleshooting

"No such file or directory"

Double-check the folder name.

Linux is case-sensitive.

`Documents` and `documents` are different.

Lost?

Run:

```
pwd
```



Then:

```
ls
```



These two commands will help you figure out where you are.

Stop and Check

You are ready to continue when:

- ☐ You can explain what a directory is.
 - ☐ You can use `pwd`.
 - ☐ You can use `ls`.
 - ☐ You can use `cd`.
 - ☐ You understand the difference between absolute and relative paths.
-

Lesson Summary

In this lesson you learned:

- Linux organizes everything into directories.
 - The root directory is `/`.
 - Your home directory is where most of your work will happen.
 - `pwd`, `ls`, and `cd` are fundamental navigation commands.
 - Absolute and relative paths describe locations in different ways.
-

What To Do Next

Continue to **Lesson 12 – Useful Commands**.

You'll build on today's navigation skills by learning several safe commands you'll use throughout the rest of the course.

2.4.4 Useful Commands

Goal

Learn beginner Linux commands that are useful in Kali.

Why this matters

Kali includes many advanced tools, but the basic Linux commands still matter.

If you can navigate, read help, and understand basic output, the advanced tools will be easier to learn later.

Plain-English explanation

Commands are typed instructions.

You do not need to memorize every command right away.

The goal is to learn a few useful commands and understand how to ask Linux for help.

New terms

- **Command:** An instruction typed into the terminal.
- **Option:** Extra information that changes how a command behaves.
- **Argument:** The thing a command works on, such as a file or folder.
- **Manual page:** Built-in help documentation for a command.
- **Process:** A running program.

Navigation commands

Show current folder:

```
pwd
```



List files:

```
ls
```



List files with details:

```
ls -l
```



Change folder:

```
cd folder-name
```



Go back one folder:

```
cd ..
```



Go to your home folder:

```
cd ~
```



File and folder commands

Create a folder:

```
mkdir practice
```



Create an empty file:

```
touch notes.txt
```



View a text file:

```
cat notes.txt
```



Copy a file:

```
cp notes.txt notes-copy.txt
```



Rename or move a file:

```
mv notes.txt new-name.txt
```



Delete a file:

```
rm new-name.txt
```



Use `rm` carefully. It deletes files.

System information commands

Show current user:

```
whoami
```



Show computer name:

```
hostname
```



Show network information:

```
ip addr
```



Show running processes:

```
ps aux
```



Help commands

Show a command manual:

```
man command-name
```



Example:

```
man ls
```



Show quick help:

```
command-name --help
```



Example:

```
nmap --help
```



Understanding command structure

Many commands follow this pattern:

```
command option target
```



Example:

```
ls -l /home/kali
```



In that example:

- `ls` is the command
- `-l` is an option
- `/home/kali` is the target

Safe beginner practice

Create a practice folder:

```
mkdir command-practice
```

Move into it:

```
cd command-practice
```

Create a notes file:

```
touch notes.txt
```

List the folder:

```
ls -l
```

Go back:

```
cd ..
```

Common mistakes

- Trying to memorize too many commands at once
- Forgetting that Linux is case-sensitive
- Running delete commands too quickly
- Not reading error messages
- Copying advanced commands without understanding them

Troubleshooting

If you do not know what a command does, try:

```
command-name --help
```

Or:

```
man command-name
```



If the manual page opens, press:

```
q
```



to quit.

Stop and check

You are ready to continue when:

- ☐ You can move between folders.
- ☐ You can create a folder.
- ☐ You can create a file.
- ☐ You understand the pattern `command option target`.
- ☐ You know how to ask a command for help.
- ☐ You understand that delete commands should be used carefully.

Summary

You learned basic commands for navigation, files, system information, and help.

These commands are the foundation for using Kali more confidently.

What to do next

Continue to Lesson 11: Networking Basics.

2.4.5 Guide 13: Finding Help

Estimated Time: 25–35 minutes

Difficulty: Beginner

Platform Support: Kali Linux

Have You Ever Wondered...

...how experienced Linux users seem to know every command?

Here's a secret:

They don't.

Professional Linux users don't memorize thousands of commands.

Instead, they know where to find trustworthy information quickly.

Learning how to find answers is one of the most valuable Linux skills you can develop.

Learning Objectives

By the end of this guide, you should be able to:

- Explain why documentation is important.
 - Use Linux manual pages.
 - Use built-in help options.
 - Search for commands by keyword.
 - Develop good habits for finding trustworthy information.
-

Goal

Learn how to find reliable answers without relying on memorization.

Why This Matters

Linux includes thousands of commands.

Nobody memorizes all of them.

Professionals regularly look up documentation, examples, and command options.

The difference between a beginner and an experienced Linux user often isn't memory—it's knowing where to look.

Plain-English Explanation

Think of Linux documentation like the owner's manual for your car.

You probably don't remember every page.

But you know where to find information when you need it.

Linux works the same way.

Instead of trying to memorize every command, you'll learn how to ask Linux for help.

New Terms

- Manual Page
 - Documentation
 - Help Option
 - Command Option
 - Keyword Search
 - Built-in Help
-

2.4.5 The Linux Manual

Linux includes built-in manuals for most commands.

To open a manual page, type:

```
man ls
```



Then press **Enter**.

You should see the manual page for the `ls` command.

You can scroll using:

- Arrow keys
- Page Up
- Page Down

To quit the manual, press:

```
q
```



Using --help

Many commands also provide a quick help screen.

Example:

```
ls --help
```



This displays a shorter explanation than the manual page.

If you only need a reminder of available options, `--help` is often the fastest choice.

Using whatis

The `whatis` command provides a short description of another command.

Example:

```
whatis pwd
```



Example output:

```
pwd (1) - print name of current working directory
```



This is useful when you only need a quick reminder.

Using apropos

What if you don't know the command name?

Use:

```
apropos keyword
```



Example:

```
apropos copy
```



Linux searches its manual pages for related commands.

This is an excellent way to discover tools you didn't know existed.

Comparing Your Options

Command	Best Used For
<code>man</code>	Full documentation
<code>--help</code>	Quick reminder
<code>whatis</code>	One-line description
<code>apropos</code>	Search by topic

Learning Tip

You do not become an expert by memorizing everything.

You become an expert by knowing how to find reliable information quickly.

Looking Under the Hood

Manual pages are installed on your computer alongside many Linux programs.

That means you can often read documentation even when you don't have an internet connection.

Why Professionals Do This

Experienced Linux users regularly read documentation before trying unfamiliar commands or options.

Spending a few minutes reading documentation can prevent mistakes and save a lot of troubleshooting later.

Beyond Built-In Help

The built-in documentation is often the best place to start.

If you still need more information, consider these sources in order:

1. Built-in help (`man` , `--help` , `whatis`)
2. Official project documentation
3. Distribution documentation (such as Kali or Debian)
4. Trusted books or tutorials
5. Community forums
6. AI assistants

When using AI, treat it as a learning partner—not as the final authority.

Always understand commands before running them.

Try It Yourself

Practice the following commands:

```
man pwd
```



Exit by pressing:

```
q
```



Next:

```
ls --help
```



Then:

```
whatis ls
```



Finally:

```
apropos network
```



Notice how each command helps you in a different way.

Reflection Questions

Without looking back:

1. Why don't professionals memorize every Linux command?
2. What does `man` do?
3. When would you use `--help` instead?
4. What does `whatis` show?
5. What is `apropos` useful for?



Memory Helper

Remember these commands:

`man command` → Read the full manual

`command --help` → Quick help

`whatis command` → Short description

`apropos keyword` → Search for commands



Common Beginner Mistake

Many beginners immediately search random websites and copy commands without understanding them.

Start with the built-in documentation first.

If you use information from the internet, prefer official documentation and trusted educational resources.



Real-World Example

Imagine you forget how to copy a file.

Instead of immediately opening a web browser, try:

```
man cp
```



or

```
cp --help
```



In many cases, you'll find the answer in seconds.



Progress, Not Perfection

Nobody memorizes every Linux command.

Focus on understanding what commands do and where to find trustworthy information.

Looking Ahead

You've learned how to find information when you're unsure about a command.

Next, you'll begin creating, copying, moving, and organizing files from the terminal.

Those are skills you'll use throughout the rest of your Linux journey.

Stop and Check

You are ready to continue when:

- ☐ You can open a manual page.
- ☐ You know how to exit a manual page.
- ☐ You understand when to use `--help`.
- ☐ You understand what `whatis` does.
- ☐ You understand what `apropos` does.
- ☐ You know where to look for trustworthy Linux documentation.

Lesson Summary

In this guide you learned:

- Professionals use documentation regularly.
 - Linux includes built-in manuals.
 - `man` opens the full manual.
 - `--help` provides quick usage information.
 - `whatis` gives a short description.
 - `apropos` searches for commands by keyword.
 - Understanding commands is more important than memorizing them.
-

What To Do Next

Continue to **Creating and Managing Files**.

You'll begin working with files and directories using safe, practical Linux commands.

2.4.6 Guide 14: Creating and Managing Files

Estimated Time: 30–40 minutes

Difficulty: Beginner

Platform Support: Kali Linux

Have You Ever Wondered...

...how people create, organize, rename, and remove files without using a graphical file manager?

Everything you can do with a mouse can also be done from the terminal.

In this guide, you'll learn how to safely create, copy, move, rename, and remove files using a few simple Linux commands.

Learning Objectives

By the end of this guide, you should be able to:

- Create directories.
 - Create empty files.
 - Copy files.
 - Move and rename files.
 - Remove files safely.
 - Remove empty directories.
 - Understand when to use each command.
-

Goal

Learn the basic file management commands you'll use throughout your Linux journey.

Why This Matters

Every operating system stores information as files and folders.

Knowing how to manage them from the terminal is an important skill for Linux users and cybersecurity professionals.

These same commands will be used throughout the rest of this learning path.

Plain-English Explanation

Think of your computer like a filing cabinet.

Sometimes you need to:

- Create a new folder.
- Create a document.
- Make a copy.
- Rename something.
- Move it somewhere else.
- Throw something away.

Linux provides simple commands for each of these tasks.

New Terms

- Directory
 - File
 - Copy
 - Move
 - Rename
 - Delete
-

2.4.6 Our Practice Project

For this guide, we'll create a practice folder and work entirely inside it.

This lets you experiment without worrying about changing important files.

2.4.6 Step 1: Create a Directory

Type:

```
mkdir practice
```



Press **Enter**.

This creates a new directory named:

```
practice
```



Think of it like creating a new folder in Windows or macOS.

2.4.6 Step 2: Enter the Directory

Type:

```
cd practice
```



You are now working inside your new directory.

Confirm your location:

```
pwd
```



2.4.6 Step 3: Create Your First File

Type:

```
touch notes.txt
```



This creates an empty file called:

```
notes.txt
```



List the contents of the directory:

```
ls
```



You should now see:

```
notes.txt
```



2.4.6 Step 4: Make a Copy

Type:

```
cp notes.txt backup.txt
```



Now list the directory again:

```
ls
```



You should see:

```
notes.txt
```



```
backup.txt
```

The original file still exists.

You now have two separate files.

2.4.6 Step 5: Rename a File

The `mv` command can also rename files.

Type:

```
mv backup.txt notes-backup.txt
```



List the directory:

```
ls
```



The file has been renamed.

2.4.6 Step 6: Move a File

Let's create another directory.

Type:

```
mkdir archive
```

Now move the backup file:

```
mv notes-backup.txt archive/
```

The file has moved into the new directory.

You can verify this by running:

```
ls
```

Then:

```
ls archive
```

2.4.6 Step 7: Remove a File

Delete the original file:

```
rm notes.txt
```

List the directory:

```
ls
```

Only the archive directory should remain.



Be Careful with rm

The `rm` command permanently removes files.

Unlike many graphical desktop environments, there is usually no Recycle Bin when using `rm`.

Before pressing **Enter**, take a moment to make sure you're deleting the correct file.

2.4.6 Step 8: Remove an Empty Directory

Return to the main directory:

```
cd archive
```

Delete the backup file:

```
rm notes-backup.txt
```

Go back:

```
cd ..
```

Now remove the empty directory:

```
rmdir archive
```

The `rmdir` command only removes directories that are already empty.

Learning Tip

Practice these commands in a temporary directory like today's `practice` folder.

Experimenting in a safe location builds confidence without risking important files.

Looking Under the Hood

Linux doesn't care whether a file was created from the terminal or from a graphical file manager.

Both methods work with the same underlying file system.

Why Professionals Do This

Many Linux administrators and cybersecurity professionals work on remote systems that don't have a graphical desktop.

Knowing these commands allows you to manage files anywhere, even over a remote connection.

Try It Yourself

Repeat the following sequence without looking back:

```
mkdir practice
cd practice
touch notes.txt
cp notes.txt backup.txt
mv backup.txt notes-backup.txt
mkdir archive
mv notes-backup.txt archive/
ls
ls archive
rm notes.txt
cd archive
rm notes-backup.txt
cd ..
rmdir archive
```

Try explaining what each command does before you press **Enter**.

Reflection Questions

Without looking back:

1. What command creates a directory?
 2. What command creates an empty file?
 3. What is the difference between `cp` and `mv`?
 4. Why should you be careful when using `rm`?
 5. Why can `rmdir` only remove empty directories?
-

✓ Memory Helper

Remember these commands:

`mkdir` → Create a directory

`touch` → Create an empty file

`cp` → Copy a file

`mv` → Move or rename a file

`rm` → Remove a file

`rmdir` → Remove an empty directory

⚠ Common Beginner Mistake

New Linux users often confuse:

`rm`

with

`rmdir`

Remember:

`rm` removes files.

`rmdir` removes empty directories.

🔧 Real-World Example

Imagine organizing photos after a vacation.

You might:

- Create a new folder.
- Copy your favorite photos.
- Rename a few pictures.
- Move them into different folders.
- Delete duplicates.

The commands you learned today perform those same tasks from the terminal.



Progress, Not Perfection

Nobody memorizes every Linux command after using it once.

Focus on understanding what each command does.

Repetition will build confidence over time.

Looking Ahead

You've learned how to create, copy, move, rename, and remove files safely.

Next, you'll learn why Linux sometimes displays "**Permission denied**" and how the Linux permission system helps protect your files.

Stop and Check

You are ready to continue when:

- ☐ You can create a directory.
- ☐ You can create a file.
- ☐ You can copy a file.
- ☐ You can rename a file.
- ☐ You can move a file.
- ☐ You understand when to use `rm`.
- ☐ You understand when to use `rmdir`.

Lesson Summary

In this guide you learned:

- How to create directories using `mkdir`.
- How to create files using `touch`.
- How to copy files using `cp`.
- How to move and rename files using `mv`.
- How to remove files using `rm`.

- How to remove empty directories using `rmdir`.

These commands form the foundation of working with files in Linux.

What To Do Next

Continue to **Understanding Permissions**.

You'll learn why Linux protects certain files and what causes the familiar "**Permission denied**" message.

2.4.7 Guide 15: Understanding Permissions

Estimated Time: 30–40 minutes

Difficulty: Beginner

Platform Support: Kali Linux

Have You Ever Wondered...

...why Linux sometimes says:

```
Permission denied
```



Even when you're the only person using the computer?

Linux protects files and folders to help prevent accidental changes, malware, and unauthorized access.

In this guide, you'll learn why permissions exist and how they help keep a Linux system secure.

Learning Objectives

By the end of this guide, you should be able to:

- Explain why Linux uses permissions.
 - Understand the concepts of users, groups, and others.
 - Read a basic permission listing.
 - Understand what `sudo` does.
 - Recognize common permission-related errors.
-

Goal

Learn why Linux protects files and how permissions help keep systems secure.

Why This Matters

Imagine if every program could change every file on your computer.

A simple mistake—or malicious software—could damage your operating system in seconds.

Linux permissions help prevent that.

They are one of the biggest reasons Linux systems are known for their stability and security.

Plain-English Explanation

Think of a building.

Some rooms are open to everyone.

Some rooms are for employees only.

Some rooms are locked.

Linux works in a similar way.

Every file and directory has rules that determine who can:

- Read it
 - Change it
 - Run it
-

New Terms

- User
 - Group
 - Owner
 - Permissions
 - Read
 - Write
 - Execute
 - Root
 - sudo
-

2.4.7 Every File Has an Owner

When a file is created, Linux records who owns it.

That owner usually has more control over the file than other users.

This helps prevent accidental or unauthorized changes.

2.4.7 Users, Groups, and Others

Linux permissions are divided into three categories:

- Owner
- Group
- Others

Think of them like this:

Category	Meaning
Owner	The person who created or owns the file
Group	A collection of users with shared access
Others	Everyone else

2.4.7 The Three Basic Permissions

Linux uses three basic permissions.

Read (r)

Allows someone to view a file.

Write (w)

Allows someone to modify a file.

Execute (x)

Allows a program or script to run.

2.4.7 Viewing Permissions

Run:

```
ls -l
```



You may see something like:

```
-rw-r--r-- 1 kali kali 125 Jul 6 notes.txt
```



Don't worry about understanding every character today.

The important idea is:

Linux is showing who can read, write, or use the file.

We'll explore this in more detail in a future learning path.

Looking Under the Hood

Every file stores permission information as part of its metadata.

When you try to open, change, or run a file, Linux checks those permissions before allowing the action.

2.4.7 What is sudo?

Some tasks require administrator privileges.

Instead of logging in as the administrator all the time, Linux allows authorized users to temporarily perform administrative tasks using:

```
sudo
```



Think of it as asking Linux:

"Please run this one command with administrator privileges."

Warning

Do not use `sudo` simply because a command fails.

First understand **why** administrator privileges are required.

Using `sudo` unnecessarily can make mistakes much more serious.

2.4.7 What Does "Permission Denied" Mean?

You may eventually see an error like:

```
Permission denied
```



This usually means one of three things:

- You don't own the file.
- You don't have permission to perform that action.
- Administrator privileges are required.

The message is protecting your system—not trying to frustrate you.

Learning Tip

When you see an error message, read it carefully.

Linux usually tells you exactly what went wrong.

Understanding the message is often the first step toward solving the problem.

Why Professionals Do This

Experienced Linux users avoid using administrator privileges unless they are actually needed.

Working as a normal user helps prevent accidental system changes and improves security.

Try It Yourself

Run:

```
ls -l
```



Look at the first column of each line.

Notice that every file has permission information.

Next, run:

```
whoami
```



Confirm which user you are currently using.

Finally, think about why Linux might restrict certain actions.

Reflection Questions

Without looking back:

1. Why does Linux use permissions?
2. What are the three permission categories?
3. What do read, write, and execute mean?
4. What does `sudo` do?
5. Why is "Permission denied" often a good thing?

✓ Memory Helper

Remember:

Owner → Usually has the most control

Group → Shared access

Others → Everyone else

Read → View

Write → Modify

Execute → Run



Common Beginner Mistake

Many beginners think:

```
Permission denied
```



means Linux is broken.

In reality, Linux is usually protecting the system from accidental or unauthorized changes.



Real-World Example

Imagine a shared office.

Employees can edit their own documents.

Managers can access additional information.

Visitors can only view public documents.

Linux permissions work in a very similar way.



Progress, Not Perfection

You are not expected to memorize Linux permissions today.

The goal is simply to understand why they exist and recognize the basic concepts.

Looking Ahead

Congratulations!

You've completed the technical lessons in this module.

Next, you'll review what you've learned and celebrate an important milestone in your Linux journey.

Stop and Check

You are ready to continue when:

- ☐ You understand why Linux uses permissions.
 - ☐ You know the difference between owner, group, and others.
 - ☐ You understand read, write, and execute permissions.
 - ☐ You know what `sudo` does.
 - ☐ You understand why "Permission denied" is often a helpful message.
-

Lesson Summary

In this guide you learned:

- Linux protects files using permissions.
 - Every file has an owner.
 - Permissions are divided into owner, group, and others.
 - Read, write, and execute determine what users can do.
 - `sudo` temporarily grants administrator privileges.
 - Permission errors usually protect the system rather than indicate a problem.
-

What To Do Next

Continue to **Module 3 Review**.

You'll review the concepts you've learned before moving on to **Understanding Networks**.

2.4.8 Module 3 Review

Congratulations!

You've completed every lesson in **Learning Your Way Around Linux**.

Take a few minutes to review what you've learned before moving on.

Remember, this isn't a test.

It's a chance to see how much you've learned and identify anything you'd like to practice again.

2.4.8 What You Learned

During this module you learned how to:

- Use the Linux terminal.
- Understand prompts and command output.
- Navigate directories.
- Understand files and folders.
- Use common Linux commands.
- Find trustworthy documentation.
- Create, move, copy, rename, and remove files.
- Understand why Linux uses permissions.

Those are foundational Linux skills used by professionals every day.

2.4.8 Quick Review

See if you can answer these without looking back.

Terminal

- What is the terminal?
 - What does the prompt tell you?
-

Navigation

What command:

- Shows your current directory?
 - Lists files?
 - Changes directories?
 - Returns to your home directory?
-

Finding Help

What command:

- Opens the manual?
 - Displays quick help?
 - Searches for commands?
-

File Management

Which command:

- Creates a directory?
 - Creates a file?
 - Copies a file?
 - Moves or renames a file?
 - Removes a file?
 - Removes an empty directory?
-

Permissions

Can you explain:

- Owner
- Group
- Others

What do:

- Read
- Write
- Execute

mean?

What does `sudo` do?

2.4.8 Practice Challenge

Without looking at previous lessons, try completing this exercise.

Create a directory named:

```
practice
```



Move into it.

Create two files.

Copy one of them.

Rename the copy.

List the directory.

Delete the copied file.

Return to your home directory.

Remove the practice directory.

Don't worry if you need to look up commands.

Knowing where to find the answer is part of learning.

✓ Memory Helper

The Linux commands you've learned so far:

`pwd`

`ls`

`cd`

`clear`

`whoami`

`mkdir`

`touch`

`cp`

`mv`

`rm`

`rmdir`

`man`

`whatis`

`apropos`

`--help`

🔥 Learning Tip

If there are commands you don't remember, that's completely normal.

Try practicing them a few minutes each day instead of trying to memorize everything at once.

🔧 Progress, Not Perfection

Every experienced Linux user once had to look up `pwd`, `ls`, and `cp`.

Confidence comes from practice, not perfection.

2.4.8 Self Check

How confident do you feel?

- ☐ I still feel overwhelmed.
- ☐ I'm starting to understand.
- ☐ I can complete basic tasks by myself.
- ☐ I'm ready for networking.

There are no right or wrong answers.

This is simply a way to reflect on your progress.

2.4.8 Looking Ahead

You've learned how to use Linux.

Now it's time to learn how computers communicate with each other.

In the next module you'll begin exploring:

- Networks
- IP addresses
- Ports
- Wi-Fi
- Ethernet
- Your own home network

These concepts are essential before using tools like Nmap and Wireshark.

What To Do Next

Continue to **Module 3 Milestone**.

2.4.9 🎉 Module 3 Milestone

Congratulations!

You have successfully completed:

2.4.9 Learning Your Way Around Linux

Take a moment to appreciate how much you've accomplished.

When you started this module, the terminal may have looked intimidating.

Now you understand how to navigate Linux, work with files, find documentation, and understand why permissions exist.

That's a significant milestone.

2.4.9 You Can Now...

- ✅ Open the terminal with confidence.
- ✅ Navigate the Linux file system.
- ✅ Understand where your files are located.
- ✅ Find built-in documentation.
- ✅ Create and manage files.
- ✅ Understand the purpose of Linux permissions.

These are the same foundational skills used by Linux users, system administrators, developers, and cybersecurity professionals every day.

2.4.9 Your Progress So Far

You have completed:

- ✅ Module 1

Introduction to Cybersecurity

✓ Module 2

Getting Started with Kali Linux

✓ Module 3

Learning Your Way Around Linux

Three major milestones are now behind you.

✓ Celebrate Your Progress

Learning Linux takes time.

Every command you practice makes the next lesson a little easier.

You've built a strong foundation that will support everything else you learn.

🔗 Progress, Not Perfection

You are not expected to remember every command.

You are expected to know where to find answers and continue practicing.

That's exactly how professionals learn.

2.4.9 What's Next?

Now the learning becomes even more exciting.

In the next module you'll learn:

- How computers communicate.
- What an IP address really is.
- Why ports exist.
- How Wi-Fi differs from Ethernet.
- How devices discover each other.
- How your home network actually works.

Once you understand networking, tools like Nmap and Wireshark will make much more sense.

2.4.9 Ready?

Continue to:

2.4.9 Module 4

Understanding Networks

You're about to learn the concepts that connect Linux to cybersecurity.

2.5 Module 4: Understanding Networks

2.5.1 Understanding Networks

Welcome!

You've completed three major milestones.

You know how to start Kali Linux.

You know your way around Linux.

Now it's time to understand how computers communicate.

This module introduces networking concepts that every cybersecurity learner should understand.

Don't worry if networking seems confusing.

We'll build your understanding one idea at a time.

2.5.1 What You'll Learn

In this module you'll learn:

- What a network actually is.
 - How devices communicate.
 - The difference between Wi-Fi and Ethernet.
 - What an IP address is.
 - Why ports exist.
 - How computers find each other.
 - How your own home network works.
-

2.5.1 Why This Matters

Nearly every cybersecurity tool depends on networking.

Before using tools like Nmap and Wireshark, you need to understand what they're actually showing you.

Think of this module as learning the roads before learning to drive.



Progress, Not Perfection

Networking can feel overwhelming at first.

That's completely normal.

We'll focus on understanding concepts instead of memorizing technical details.

2.5.1 Module Roadmap

In this module you'll complete:

1. What Is a Network?
2. Wi-Fi vs Ethernet
3. Understanding IP Addresses
4. Understanding Ports
5. How Devices Find Each Other
6. Understanding Your Home Network

2.5.1 By the End of This Module

You'll be able to explain:

- How computers communicate.
- Why every device has an address.
- What routers actually do.
- Why ports exist.
- Why DNS is important.
- How your home network is organized.

These concepts are essential before beginning your first network scans.

2.5.1 Let's Get Started

Continue to **What Is a Network?**

2.5.2 Guide 16: What Is a Network?

Estimated Time: 25–35 minutes

Difficulty: Beginner

Have You Ever Wondered...

...how your phone can send a picture to another phone?

Or how your laptop can load a website that's stored somewhere else in the world?

The answer is simple:

Networks.

Learning Objectives

By the end of this guide you'll be able to:

- Explain what a network is.
 - Understand why networks exist.
 - Recognize examples of networks.
 - Explain the difference between local and global networks.
-

Goal

Understand the basic purpose of computer networks.

Why This Matters

Cybersecurity is largely about protecting communication.

Before we can understand attacks or defenses, we need to understand what is being protected.

That starts with understanding networks.

Plain-English Explanation

A network is simply two or more devices that can communicate with each other.

That's it.

The devices might be:

- Computers
- Phones
- Tablets
- Printers
- TVs
- Cameras
- Game consoles
- Servers

If they can exchange information, they're part of a network.

New Terms

- Network
 - Device
 - Local Area Network (LAN)
 - Internet
-

2.5.2 Why Networks Exist

Imagine having one computer with no network.

You couldn't:

- Browse websites.
- Watch videos.
- Send email.
- Print to a network printer.

- Play online games.
- Share files.

Networks allow devices to share information.

2.5.2 Local Networks

A network inside your home is called a:

Local Area Network

or

LAN.

Your router usually connects everything together.

2.5.2 The Internet

The Internet is simply an enormous collection of networks connected together.

Your home network connects to your Internet Service Provider.

Your provider connects to many other networks.

Eventually your request reaches the destination you're trying to contact.

Learning Tip

Don't think of the Internet as one giant computer.

Think of it as millions of smaller networks connected together.

Looking Under the Hood

Data doesn't travel all at once.

It is broken into smaller pieces called packets.

We'll learn more about packets later when we use Wireshark.



Why Professionals Do This

Network professionals often start troubleshooting by asking one question:

"Can these two devices communicate?"

That simple question solves many networking problems.

Try It Yourself

Look around the room.

Write down every device connected to your home network.

Examples:

- Phone
- Laptop
- Smart TV
- Gaming console
- Printer
- Tablet

You might be surprised how many devices you find.

Reflection Questions

1. What is a network?
 2. What is a LAN?
 3. What is the Internet?
 4. Why do networks exist?
 5. Can a printer be part of a network?
-

✓ Memory Helper

Network

→ Devices communicating

LAN

→ Local network

Internet

→ Networks connected together

⚠ Common Beginner Mistake

Many people think Wi-Fi is the network.

Wi-Fi is simply one way devices connect to a network.

🏠 Real-World Example

Imagine your family sharing a printer.

Instead of copying files to a USB drive every time someone wants to print, everyone sends documents across the network.

📝 Progress, Not Perfection

If networking feels unfamiliar, that's okay.

By the end of this module, you'll understand how the pieces fit together.

Looking Ahead

Now that you know what a network is...

Let's look at the two most common ways devices connect:

- Wi-Fi

- Ethernet
-

Stop and Check

You are ready to continue when:

- ☐ You can explain what a network is.
 - ☐ You understand what a LAN is.
 - ☐ You understand that the Internet is many networks connected together.
 - ☐ You can identify devices on your own home network.
-

Lesson Summary

In this guide you learned:

- A network is simply devices communicating.
 - Home networks are usually LANs.
 - The Internet connects many networks together.
 - Most modern devices participate in networks.
-

What To Do Next

Continue to **Wi-Fi vs Ethernet**.

2.5.3 Guide 17: Wi-Fi vs Ethernet

Estimated Time: 25–35 minutes

Difficulty: Beginner

Have You Ever Wondered...

...why some devices use Wi-Fi while others use an Ethernet cable?

Do they work differently?

Is one more secure than the other?

The answer is that both connect devices to the same network—they simply use different methods.

Learning Objectives

By the end of this guide, you should be able to:

- Explain the difference between Wi-Fi and Ethernet.
 - Describe the advantages of each.
 - Understand when each connection type is commonly used.
 - Recognize that both become part of the same local network.
-

Goal

Understand the two most common ways devices connect to a network.

Why This Matters

Before learning about IP addresses and network tools, it's important to understand how devices physically connect to a network.

Whether a device uses Wi-Fi or Ethernet, it still communicates using the same networking concepts.

Plain-English Explanation

Imagine your home network as a highway.

Every device wants to reach the same destination.

Some vehicles use one entrance.

Others use another.

They all end up on the same highway.

Wi-Fi and Ethernet work the same way.

They simply provide different paths to the same network.

New Terms

- Wi-Fi
 - Ethernet
 - Wireless
 - Wired
 - Router
 - Switch
-

2.5.3 Wi-Fi

Wi-Fi connects devices without a cable.

Examples include:

- Phones
- Tablets
- Laptops
- Smart speakers
- Smart TVs

Advantages:

- Mobility
- Convenience

- Easy to connect

Disadvantages:

- Can experience interference.
 - Speed may vary.
 - Signal strength changes with distance and obstacles.
-

2.5.3 Ethernet

Ethernet uses a physical network cable.

Examples include:

- Desktop computers
- Servers
- Gaming consoles
- Network switches
- Some printers

Advantages:

- Stable connection
- Low latency
- Consistent speed
- Less susceptible to wireless interference

Disadvantages:

- Requires a cable.
 - Limits how far a device can move.
-

2.5.3 Are They Different Networks?

No.

If both are connected to the same router, they're usually part of the same local network.

For example:

- Your phone may use Wi-Fi.
- Your desktop computer may use Ethernet.

They can still communicate with each other.

Which One Is Better?

There isn't a universal answer.

Choose the connection that best fits the situation.

Situation	Usually Best
Phone	Wi-Fi
Tablet	Wi-Fi
Laptop around the house	Wi-Fi
Desktop computer	Ethernet
Home server	Ethernet
Gaming PC	Ethernet (when practical)

Learning Tip

Don't think of Wi-Fi and Ethernet as competitors.

Think of them as two different ways to reach the same network.

Looking Under the Hood

Once a device joins the network, higher-level networking concepts such as IP addresses, DNS, and ports work the same regardless of whether the connection is wired or wireless.



Why Professionals Do This

Network administrators often use Ethernet for infrastructure such as servers and switches because it provides a stable, predictable connection.

Wi-Fi is then used where mobility and convenience are more important.

Try It Yourself

Look around your home.

Create two lists.

Wi-Fi Devices

Write down every device connected wirelessly.

Ethernet Devices

Write down every device connected with a network cable.

Think about why each device uses that connection type.

Reflection Questions

1. What is the main difference between Wi-Fi and Ethernet?
 2. Which connection type uses cables?
 3. Which connection type allows greater mobility?
 4. Can Wi-Fi and Ethernet devices communicate with each other?
 5. Why might a server use Ethernet instead of Wi-Fi?
-

✓ Memory Helper

Wi-Fi

→ Wireless connection

Ethernet

→ Wired connection

Both

→ Connect devices to the same network

⚠ Common Beginner Mistake

Many people think Wi-Fi and Ethernet are separate networks.

In most homes, they are simply different ways to connect to the same local network.

💡 Real-World Example

Your laptop may connect using Wi-Fi while your desktop computer uses Ethernet.

Even though they connect differently, they can still share files, print to the same printer, and communicate with each other.

🔧 Progress, Not Perfection

You don't need to memorize networking standards or cable categories today.

Focus on understanding the purpose of each connection type.

Looking Ahead

Now that you understand how devices connect to a network, it's time to answer another important question:

How does every device know where to send information?

The answer begins with **IP addresses**.

Stop and Check

You are ready to continue when:

- ☐ You understand the difference between Wi-Fi and Ethernet.
 - ☐ You know when each is commonly used.
 - ☐ You understand that both can belong to the same local network.
 - ☐ You can identify which devices in your home use each connection type.
-

Lesson Summary

In this guide you learned:

- Wi-Fi provides a wireless network connection.
 - Ethernet provides a wired network connection.
 - Both connect devices to the same local network.
 - Different situations benefit from different connection types.
-

What To Do Next

Continue to **Understanding IP Addresses**.

2.5.4 Guide 18: Understanding IP Addresses

Estimated Time: 30–40 minutes

Difficulty: Beginner

Have You Ever Wondered...

...how your computer knows where to send information?

When you open a website, send a message, or print to a network printer, your device needs to know where that information should go.

That's where IP addresses come in.

Learning Objectives

By the end of this guide, you should be able to:

- Explain what an IP address is.
 - Understand why every network device needs an address.
 - Recognize the difference between public and private IP addresses.
 - Identify examples of common home network addresses.
-

Goal

Understand the purpose of IP addresses and how they allow devices to communicate.

Why This Matters

Imagine trying to mail a letter without knowing the destination address.

It would never arrive.

Networks work the same way.

Every device needs an address so information knows where to go.

Plain-English Explanation

Think of an IP address like a street address.

Your house has an address so people know where to deliver mail.

Your computer has an IP address so other devices know where to send information.

Without addresses, devices wouldn't know who should receive the data.

New Terms

- IP Address
 - IPv4
 - Public IP Address
 - Private IP Address
 - Router
-

2.5.4 Every Device Needs an Address

Every device connected to a network receives an IP address.

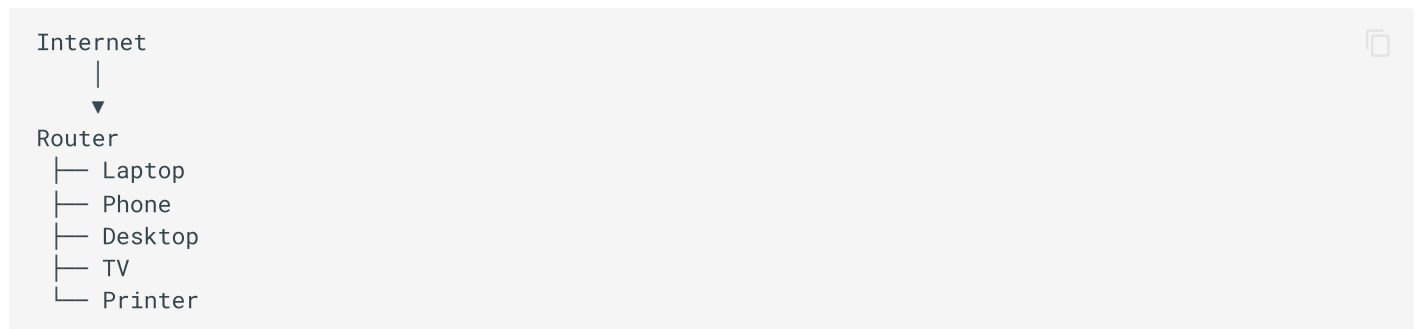
Examples include:

- Desktop computers
- Laptops
- Phones
- Tablets
- Smart TVs
- Printers
- Cameras

If a device communicates on the network, it needs an address.

2.5.4 A Typical Home Network

A home network often looks something like this:



Each of those devices has its own IP address.

2.5.4 IPv4 Addresses

Most home networks still use IPv4 addresses.

They look like this:

```
192.168.1.15
```

An IPv4 address contains four numbers separated by periods.

Other common examples include:

```
192.168.0.100
10.0.0.50
172.16.1.25
```

Don't worry about memorizing these today.

Just recognize the pattern.

2.5.4 Private IP Addresses

Devices inside your home network usually use private IP addresses.

These addresses are only used inside your local network.

Your router keeps track of them.

2.5.4 Public IP Addresses

Your Internet Service Provider assigns your home a public IP address.

This is the address websites see when you visit them.

Think of it as the address for your entire house.

Inside the house, each room has its own location.

Outside the house, everyone uses the same street address.

Public vs Private

Private IP	Public IP
Used inside your network	Used on the Internet
Assigned by your router	Assigned by your ISP
Not directly reachable from the Internet	Used to communicate with other networks

Learning Tip

Don't worry about memorizing address ranges.
Focus on understanding **why** addresses exist.

Looking Under the Hood

Your router keeps track of which local device requested information from the Internet and sends the replies back to the correct device.



Why Professionals Do This

One of the first things a network administrator checks is a device's IP address.

If a device doesn't have a valid address, it usually won't be able to communicate properly.

Try It Yourself

Open a terminal and type:

```
ip addr
```



Look for an address assigned to your primary network connection.

You may also use:

```
hostname -I
```



to display your device's IP address.

Don't worry if you see more than one address.

That can happen on some systems.

Reflection Questions

1. Why does every network device need an IP address?
2. What is the difference between a public IP address and a private IP address?
3. Who usually assigns your private IP address?
4. Who usually assigns your public IP address?
5. Can two devices on the same network use the same IP address?

✓ Memory Helper

IP Address

→ Identifies a device

Private IP

→ Used inside your network

Public IP

→ Used on the Internet

Router

→ Connects your network to the Internet

⚠ Common Beginner Mistake

Many beginners think every device has a public IP address.

In most homes, devices use private IP addresses while the router communicates with the Internet using a public IP address.

🏠 Real-World Example

Imagine an apartment building.

The building has one street address.

Each apartment has its own number.

A public IP address is like the building's address.

Private IP addresses are like the apartment numbers.

📝 Progress, Not Perfection

You don't need to memorize IP address ranges today.

Understanding their purpose is much more important.

Looking Ahead

Now that every device has an address...

How does one device know which application should receive the information?

The answer is **ports**.

Stop and Check

You are ready to continue when:

- ☐ You know what an IP address is.
 - ☐ You understand why every device needs one.
 - ☐ You know the difference between public and private IP addresses.
 - ☐ You understand the router's role in assigning local addresses.
-

Lesson Summary

In this guide you learned:

- Every device needs an IP address.
 - IP addresses identify devices on a network.
 - Home networks use private IP addresses.
 - Internet communication uses a public IP address.
 - Routers connect local devices to the Internet.
-

What To Do Next

Continue to **Understanding Ports**.

2.5.5 Guide 19: Understanding Ports

Estimated Time: 30–40 minutes

Difficulty: Beginner

Have You Ever Wondered...

...how your computer knows whether incoming information belongs to your web browser, your email program, or an online game?

An IP address gets information to the correct device.

Ports help deliver that information to the correct application on that device.

Learning Objectives

By the end of this guide, you should be able to:

- Explain what a network port is.
 - Understand why ports exist.
 - Recognize several common port numbers.
 - Understand the relationship between IP addresses and ports.
-

Goal

Understand how ports help computers organize network communication.

Why This Matters

An IP address tells information which computer to reach.

A port tells it where to go once it gets there.

Without ports, computers couldn't run multiple network applications at the same time.

Plain-English Explanation

Imagine an apartment building.

The street address gets your package to the building.

The apartment number tells the delivery driver which apartment should receive it.

Networking works the same way.

IP Address

↓

Correct device

Port

↓

Correct application

New Terms

- Port
 - Service
 - TCP
 - UDP
-

2.5.5 What Is a Port?

A network port is simply a numbered communication endpoint.

Think of it like a doorway for a specific application.

Different applications listen on different ports.

2.5.5 Why Ports Exist

Imagine your computer is running:

- A web browser
- Email
- Discord
- Steam
- Spotify

All of these applications communicate over the network.

Ports help Linux (and other operating systems) know which application should receive each piece of information.

2.5.5 Common Port Numbers

Here are a few examples.

Port	Common Service
22	SSH
53	DNS
80	HTTP
443	HTTPS

Don't worry about memorizing these today.

You'll encounter them naturally as you continue learning.

2.5.5 IP Address + Port

Think of it this way.

IP Address

↓

Find the computer.

Port



Find the application.

Both are needed for successful communication.

TCP and UDP

Most network communication uses one of two protocols:

TCP

TCP focuses on reliable communication.

If information is lost, TCP asks for it again.

Examples include:

- Websites
 - Email
 - File transfers
-

UDP

UDP focuses on speed.

It does not wait for missing information.

Examples include:

- Voice calls
- Video streaming
- Online gaming

For now, simply remember that both are common ways for applications to communicate.

Learning Tip

Don't try to memorize dozens of port numbers.

Learn what ports are first.

The commonly used ports will become familiar naturally through practice.

Looking Under the Hood

Applications "listen" on ports.

When information arrives for a particular port, the operating system delivers it to the application that's listening there.

Why Professionals Do This

Network administrators often look at open ports to understand what services are running on a computer.

Cybersecurity professionals do the same thing during authorized security assessments.

Try It Yourself

Think about the applications running on your computer.

Examples might include:

- Web browser
- Music player
- Discord
- Steam
- Email client

Every network-enabled application uses one or more ports when communicating.

You don't need to identify the port numbers yet.

Just recognize that each application communicates differently.

Reflection Questions

1. What does an IP address identify?
2. What does a port identify?
3. Why do computers need ports?
4. Name one common port number.
5. What is one difference between TCP and UDP?

✓ Memory Helper

IP Address

→ Find the device

Port

→ Find the application

TCP

→ Reliable communication

UDP

→ Fast communication

⚠ Common Beginner Mistake

Beginners often think ports are physical connectors.

Network ports are numbers used by software, not the USB or Ethernet ports on your computer.

🔧 Real-World Example

Imagine calling a large company.

The company's main phone number gets you to the building.

An extension number connects you to a specific employee.

IP addresses are like the main phone number.

Ports are like the extension numbers.



Progress, Not Perfection

Very few professionals memorize every port number.

Understanding why ports exist is much more important than memorizing numbers.

Looking Ahead

Now you understand:

- Devices have IP addresses.
- Applications use ports.

Next you'll learn how computers find websites using names instead of numbers.

Stop and Check

You are ready to continue when:

- ☐ You know what a port is.
 - ☐ You understand why ports exist.
 - ☐ You know the difference between an IP address and a port.
 - ☐ You recognize a few common port numbers.
 - ☐ You understand the basic idea of TCP and UDP.
-

Lesson Summary

In this guide you learned:

- Ports identify applications running on a device.
 - IP addresses identify devices.
 - Computers use both addresses and ports for communication.
 - TCP emphasizes reliability.
 - UDP emphasizes speed.
-

What To Do Next

Continue to **How Devices Find Each Other (DNS)**.

2.5.6 Guide 20: How Devices Find Each Other (DNS)

Estimated Time: 30–40 minutes

Difficulty: Beginner

Have You Ever Wondered...

...why you can type:

```
openai.com
```



instead of something like:

```
172.64.154.211
```



Most people don't memorize IP addresses.

Instead, computers use a system called the **Domain Name System (DNS)** to translate easy-to-remember names into IP addresses.

Learning Objectives

By the end of this guide, you should be able to:

- Explain what DNS is.
 - Understand why DNS exists.
 - Describe what happens when you visit a website.
 - Recognize the relationship between domain names and IP addresses.
-

Goal

Understand how computers find websites using names instead of numbers.

Why This Matters

Imagine trying to remember the phone number of every person you've ever called.

Most people don't.

Instead, they save names in their contacts.

DNS works in much the same way.

Instead of remembering IP addresses, we remember names.

Plain-English Explanation

Computers communicate using IP addresses.

People prefer names.

DNS connects the two.

When you type:

`example.com`



your computer asks a DNS server:

"What IP address belongs to this name?"

The DNS server replies with the correct IP address.

Your computer can then communicate with the website.

New Terms

- Domain Name
 - DNS
 - DNS Server
 - Hostname
 - Resolution
-

2.5.6 A Simple Example

Imagine you want to visit:

```
openai.com
```



Your computer does **not** automatically know where that website is.

Instead, it asks a DNS server:

```
"What's the IP address for openai.com?"
```

The DNS server responds with the correct address.

Only then does your computer begin communicating with the website.

2.5.6 Why Not Just Use IP Addresses?

You certainly can.

But imagine trying to remember hundreds of addresses like:

```
203.0.113.42
```



instead of:

```
example.com
```



Names are easier for people.

Numbers are easier for computers.

DNS allows both to work together.

2.5.6 Where Does DNS Come From?

Your device usually learns which DNS server to use from your:

- Home router
- Internet Service Provider (ISP)
- Manual configuration (less common for beginners)

You don't need to configure DNS yourself to understand how it works.

Learning Tip

Think of DNS as the Internet's contact list.

You remember names.

Your computer looks up the numbers.

Looking Under the Hood

DNS requests usually happen very quickly.

Most of the time, you don't even notice they're happening before a website loads.

Why Professionals Do This

Network administrators often check DNS first when a website won't load.

Sometimes the website is working perfectly—the problem is simply that the name isn't resolving correctly.

Try It Yourself

Open a terminal and type:

```
ping openai.com
```



(You don't need to let it run for long. Press **Ctrl + C** to stop it.)

Notice that the command displays an IP address before sending any network traffic.

That's DNS in action.

If `ping` isn't available or is blocked on your system, don't worry. We'll explore more networking tools later in the course.

Reflection Questions

Without looking back:

1. What does DNS stand for?

2. Why does DNS exist?
3. What is the difference between a domain name and an IP address?
4. Why do people prefer names?
5. Why do computers still need IP addresses?

✓ Memory Helper

Domain Name

→ Easy for people to remember

IP Address

→ Used by computers

DNS

→ Translates names into IP addresses

⚠ Common Beginner Mistake

Many beginners think a website's name is the same thing as its IP address.

They are different.

DNS connects the two.

📞 Real-World Example

Think about your phone's contacts.

You tap:

Mom

instead of remembering a ten-digit phone number.

DNS performs a similar job for computers.



Progress, Not Perfection

You don't need to understand every detail of DNS today.

The important idea is simple:

Names are for people.

IP addresses are for computers.

Looking Ahead

You've learned:

- Devices have IP addresses.
- Applications use ports.
- DNS helps computers find one another using names.

Next, we'll put everything together by exploring your own home network and seeing how all of these pieces work together.

Stop and Check

You are ready to continue when:

- ☐ You understand what DNS does.
 - ☐ You know why domain names exist.
 - ☐ You understand that computers still communicate using IP addresses.
 - ☐ You can explain DNS in your own words.
-

Lesson Summary

In this guide you learned:

- DNS stands for Domain Name System.
- DNS translates domain names into IP addresses.
- People remember names more easily than numbers.
- Computers still rely on IP addresses to communicate.

- DNS makes the Internet much easier to use.
-

What To Do Next

Continue to **Understanding Your Home Network**.

You'll bring together everything you've learned in this module and see how the devices in your own home communicate with one another.

2.5.7 Guide 21: Understanding Your Home Network

Estimated Time: 35–45 minutes

Difficulty: Beginner

Have You Ever Wondered...

...what actually happens when you open a website from your laptop?

Your laptop doesn't connect directly to the Internet.

Instead, several pieces work together to make that connection possible.

In this guide, you'll bring together everything you've learned throughout this module.

Learning Objectives

By the end of this guide, you should be able to:

- Describe the major parts of a home network.
 - Explain the role of a router.
 - Understand how devices communicate on a local network.
 - Explain what happens when visiting a website.
 - See how networking concepts fit together.
-

Goal

Understand how the devices in your home work together to communicate with each other and with the Internet.

Why This Matters

Every cybersecurity professional begins by understanding the environment they're working in.

Before using security tools, it's important to understand the normal behavior of a home network.

Once you understand what's normal, unusual behavior becomes much easier to recognize.

Plain-English Explanation

Think of your home network as a neighborhood.

Each house has an address.

Roads connect the houses.

A central road connects the neighborhood to the rest of the city.

Your home network works in a very similar way.

Devices communicate with one another through your router, which connects your local network to the Internet.

New Terms

- Router
 - Local Network (LAN)
 - Internet
 - ISP
 - Device
 - Traffic
-

2.5.7 A Typical Home Network

Imagine your home contains:

- Laptop
- Desktop computer
- Phone
- Tablet
- Smart TV
- Printer
- Security camera
- Gaming console

Every one of these devices communicates across your local network.

Some use Wi-Fi.

Others use Ethernet.

They all connect through the router.

2.5.7 The Router

Your router is the center of your home network.

It helps devices:

- Communicate with one another.
- Reach the Internet.
- Receive local IP addresses.
- Send and receive network traffic.

Think of it as a traffic coordinator.

It doesn't create the information—it helps move it where it needs to go.

2.5.7 Visiting a Website

Let's put everything together.

Imagine you open your browser and visit:

example.com



Here's what happens:

1. Your computer asks DNS for the website's IP address.
2. DNS replies with the correct IP address.
3. Your computer sends the request toward your router.
4. Your router forwards the request to the Internet.
5. The website responds.
6. The response travels back through your router.
7. Your browser displays the page.

All of this usually happens in less than a second.

2.5.7 Everything Working Together

Let's review the pieces you've learned.

Component	Purpose
Network	Allows devices to communicate
Wi-Fi / Ethernet	Connect devices to the network
IP Address	Identifies a device
Port	Identifies an application
DNS	Finds IP addresses using names
Router	Connects your network to other networks

Notice how each concept builds on the previous one.

Learning Tip

When learning networking, always ask yourself:

"Where is the information now?"

Following the path of the information often makes networking much easier to understand.

Looking Under the Hood

Every time you browse the Internet, thousands of small pieces of information travel back and forth between your device and many different systems.

Networking tools allow professionals to observe and understand that communication.



Why Professionals Do This

Before troubleshooting a network problem, experienced professionals first develop a clear picture of how the network is supposed to work.

Understanding the normal flow of communication makes unusual behavior much easier to identify.

Try It Yourself

Look around your home.

Make a list of every network-connected device you can find.

For each device, ask yourself:

- Does it use Wi-Fi or Ethernet?
- Does it have an IP address?
- Does it communicate with other devices?
- Does it need the Internet, or only the local network?

You may discover more connected devices than you expected.

Reflection Questions

Without looking back:

1. What is the router's primary job?
2. What happens before your browser contacts a website?
3. Why does every device need an IP address?
4. Why do applications use ports?
5. How do all of these concepts work together?

✓ Memory Helper

Network

→ Devices communicating

Wi-Fi / Ethernet

→ Connection methods

IP Address

→ Device identifier

Port

→ Application identifier

DNS

→ Name lookup

Router

→ Connects networks

⚠ Common Beginner Mistake

Beginners often think the router is the Internet.

It isn't.

Your router connects your local network to your Internet Service Provider, which then connects you to the broader Internet.

🔧 Real-World Example

Imagine sending a letter.

You write a person's name on the envelope.

The postal service figures out where it needs to go.

The roads carry it to the destination.

Networking works similarly.

Every component has a specific job, and they all work together to deliver information.



Progress, Not Perfection

You don't need to understand every networking technology today.

The important thing is understanding how the major pieces fit together.

Looking Ahead

Congratulations!

You've completed the technical lessons in this module.

Next, you'll review everything you've learned before beginning your first cybersecurity tools.

Stop and Check

You are ready to continue when:

- ☐ You understand the role of a router.
 - ☐ You understand how devices communicate on your home network.
 - ☐ You understand how DNS, IP addresses, and ports work together.
 - ☐ You can describe what happens when you visit a website.
-

Lesson Summary

In this guide you learned:

- The router connects your local network to other networks.
 - Devices communicate using IP addresses.
 - Applications communicate using ports.
 - DNS translates names into IP addresses.
 - All of these components work together to make modern networking possible.
-

What To Do Next

Continue to **Module 4 Review**.

2.5.8 Module 4 Review

Congratulations!

You've completed **Understanding Networks**.

Networking can seem intimidating at first, but you've already learned the concepts that form the foundation of modern computer networking.

Take a few minutes to review what you've learned before moving on.

2.5.8 What You Learned

During this module you learned:

- What a network is.
- How devices connect using Wi-Fi and Ethernet.
- Why every device needs an IP address.
- What network ports are.
- How DNS translates names into IP addresses.
- How your home network works.

These concepts are used every day by system administrators, network engineers, developers, and cybersecurity professionals.

2.5.8 Quick Review

Without looking back...

Networks

- What is a network?
 - What is a LAN?
 - What is the Internet?
-

Connections

- What is the difference between Wi-Fi and Ethernet?
 - Can devices using different connection types still communicate?
-

IP Addresses

- Why does every device need an IP address?
 - What is the difference between a public IP and a private IP?
-

Ports

- What does an IP address identify?
 - What does a port identify?
 - Why are both needed?
-

DNS

- Why does DNS exist?
 - What happens when you type a website name into your browser?
-

Home Network

Can you explain the job of your:

- Router
- Laptop
- Phone
- DNS
- Internet connection

How do they all work together?

2.5.8 Practice Challenge

Look around your home.

Choose five network-connected devices.

For each one, answer:

- Does it use Wi-Fi or Ethernet?
- Does it have an IP address?
- Does it communicate with the Internet?
- What might it use the network for?

You don't need exact technical answers.

Focus on thinking through how the devices communicate.

Memory Helper

Network

→ Devices communicating

Wi-Fi / Ethernet

→ Connection methods

IP Address

→ Device

Port

→ Application

DNS

→ Name lookup

Router

→ Connects networks

Learning Tip

Networking becomes much easier once you begin thinking about where information is traveling instead of trying to memorize terminology.



Progress, Not Perfection

You don't need to understand every networking protocol today.

You've built a strong foundation that future lessons will continue to strengthen.

2.5.8 Self Check

How confident do you feel?

- ☐ Networking still feels confusing.
- ☐ I'm beginning to understand how devices communicate.
- ☐ I can explain a basic home network.
- ☐ I'm ready to begin using networking tools.

There are no right or wrong answers.

This review is for your own reflection.

2.5.8 Looking Ahead

In the next module you'll begin using your first cybersecurity tools.

You'll learn what they do, how they work, and—most importantly—how to use them responsibly.

What To Do Next

Continue to **Module 4 Milestone**.

2.5.9 🎉 Module 4 Milestone

Congratulations!

You've successfully completed:

2.5.9 Understanding Networks

Take a moment to appreciate how much you've learned.

Networking is one of the most important foundations in cybersecurity.

You now understand concepts that many people use every day without ever thinking about how they work.

2.5.9 You Can Now...

- ✅ Explain what a network is.
- ✅ Understand the difference between Wi-Fi and Ethernet.
- ✅ Explain why devices need IP addresses.
- ✅ Understand the purpose of network ports.
- ✅ Explain how DNS works.
- ✅ Describe how a typical home network communicates.

These are the same concepts you'll use when exploring cybersecurity tools.

2.5.9 Your Progress So Far

You have completed:

- ✅ Module 1

Introduction to Cybersecurity

- ✅ Module 2

Getting Started with Kali Linux

- ✅ Module 3

✓ Module 4

Understanding Networks

You have now completed the foundational portion of this learning path.

✓ **Celebrate Your Progress**

You've built the knowledge needed to understand what cybersecurity tools are actually showing you.

That's an important milestone.

🔗 **Progress, Not Perfection**

You don't need to memorize networking details.

Understanding how the pieces fit together is far more valuable.

2.5.9 What's Next?

Now the fun begins.

In Module 5 you'll begin exploring your first cybersecurity tools.

You'll learn:

- Why Kali includes so many tools.
- What Nmap does.
- What Wireshark does.
- How to safely explore your own network.
- How to begin thinking like a cybersecurity professional.

Remember:

Tools don't make someone good at cybersecurity.

Understanding does.

2.5.9 Ready?

Continue to:

2.5.9 Module 5

Your First Security Tools

2.6 Module 5: Your First Security Tools

2.6.1 Your First Security Tools

Welcome!

You've built a strong foundation.

You understand Linux.

You understand networking.

Now it's finally time to explore some of the tools that make Kali Linux unique.

Before we begin, remember something important:

Tools don't make someone good at cybersecurity.

Understanding does.

These tools simply help us observe and understand computer systems and networks.

Throughout this module, you'll only use tools in safe, ethical, and responsible ways.

2.6.1 What You'll Learn

In this module you'll learn:

- Why Kali includes so many tools.
 - What Nmap is designed to do.
 - What Wireshark is used for.
 - How to safely explore your own network.
 - How to document what you discover.
-

2.6.1 Why This Matters

Cybersecurity tools don't magically find problems.

They collect information.

Your understanding allows you to interpret that information correctly.

The stronger your foundation, the more valuable these tools become.



Progress, Not Perfection

You are not expected to become an expert in Nmap or Wireshark during this module.

The goal is simply to become comfortable with what these tools are, what they do, and how to use them responsibly.

2.6.1 Module Roadmap

In this module you'll complete:

1. Why Kali Includes So Many Tools
2. Meeting Nmap
3. Meeting Wireshark
4. Safe Network Discovery
5. Your First Home Network Inventory

2.6.1 By the End of This Module

You'll be able to:

- Explain the purpose of several common cybersecurity tools.
- Understand what Nmap and Wireshark are designed to do.
- Safely examine your own network.
- Begin documenting your observations.
- Continue learning with confidence.

2.6.1 Ethics First

Cybersecurity is built on trust.

Only use the techniques and tools you learn in this course on:

- Systems you own
- Systems you have explicit permission to test
- Authorized learning environments

Never assume permission.

Always ask first.

2.6.1 Let's Get Started

Continue to **Why Kali Includes So Many Tools**.

2.6.2 Guide 22: Why Kali Includes So Many Tools

Estimated Time: 25–35 minutes

Difficulty: Beginner

Have You Ever Wondered...

...why Kali Linux comes with hundreds of tools?

Do cybersecurity professionals actually use all of them?

The short answer is:

No.

Kali includes tools for many different specialties, and no one is expected to master them all.

Learning Objectives

By the end of this guide, you should be able to:

- Explain why Kali includes so many tools.
 - Understand that different tools solve different problems.
 - Recognize several categories of cybersecurity tools.
 - Understand why learning concepts is more important than collecting tools.
-

Goal

Understand why Kali contains so many tools and how professionals choose the right one for a particular task.

Why This Matters

New learners often open Kali for the first time and feel overwhelmed.

Seeing hundreds of tools can make it seem like cybersecurity requires memorizing everything.

It doesn't.

Professionals learn tools gradually as they encounter new problems.

Plain-English Explanation

Think of a large mechanic's toolbox.

It may contain hundreds of tools.

A mechanic doesn't use every tool on every repair.

Instead, they choose the right tool for the job.

Kali works the same way.

New Terms

- Tool
 - Framework
 - Reconnaissance
 - Analysis
 - Forensics
-

2.6.2 Why So Many Tools?

Cybersecurity includes many specialties.

Examples include:

- Network analysis
- Digital forensics
- Web application testing
- Wireless security
- Password auditing
- Malware analysis
- Incident response

Each specialty requires different tools.

2.6.2 You Don't Need to Learn Them All

One of the biggest myths about cybersecurity is:

■ "Experts know every tool."

They don't.

Experienced professionals understand concepts first.

Then they learn the tools that support their work.

2.6.2 The Right Tool for the Job

Imagine trying to tighten a screw using a hammer.

The hammer isn't a bad tool.

It's simply the wrong one for that task.

Cybersecurity works the same way.

Learning Tip

Learn concepts first.

Learn tools second.

Tools change over time.

Strong fundamentals remain valuable for years.

Looking Under the Hood

Many tools included with Kali Linux are open-source projects created and maintained by different communities around the world.

Why Professionals Do This

Experienced professionals often use only a relatively small collection of tools regularly.

They expand their toolbox as new challenges require it.

Try It Yourself

Open the Kali Applications menu.

Browse the available tool categories.

Don't worry about understanding every tool.

Simply notice how many different specialties exist.

Reflection Questions

1. Why does Kali include so many tools?
2. Do professionals use every tool?
3. Why are concepts more important than tools?
4. Can one tool solve every cybersecurity problem?
5. What surprised you most about Kali's tool collection?

Memory Helper

Tools

→ Solve specific problems

Concepts

→ Last much longer

Learn concepts first.

Learn tools second.



Common Beginner Mistake

Many beginners believe they need to master every tool in Kali.

That's neither realistic nor necessary.



Real-World Example

A carpenter owns many tools.

They don't use every tool every day.

They choose the one that best fits the task.

Cybersecurity professionals do the same thing.



Progress, Not Perfection

Nobody becomes an expert by memorizing hundreds of tools.

Focus on understanding what each tool is designed to accomplish.

Looking Ahead

Next you'll meet one of the best-known cybersecurity tools:

Nmap

You'll learn what it does, why it's useful, and how to use it responsibly on networks you own or are authorized to assess.

Stop and Check

You are ready to continue when:

- ☐ You understand why Kali contains many tools.
- ☐ You understand that no one uses every tool.
- ☐ You understand the importance of learning concepts before tools.
- ☐ You know that different tools solve different problems.

Guide Summary

In this guide you learned:

- Kali includes tools for many cybersecurity specialties.
 - Different tools solve different problems.
 - Professionals learn tools gradually.
 - Concepts are more valuable than memorizing tool names.
-

What To Do Next

Continue to **Meeting Nmap**.

2.6.3 Guide 23: Meeting Nmap

Estimated Time: 35–45 minutes

Difficulty: Beginner

Have You Ever Wondered...

...how a cybersecurity professional can discover what devices are connected to a network?

Or how they determine which services a computer is offering?

One of the most widely used tools for answering those questions is **Nmap**.

Today, you'll meet Nmap and learn what it is designed to do.

Learning Objectives

By the end of this guide, you should be able to:

- Explain what Nmap is.
 - Understand what Nmap is designed to do.
 - Recognize common situations where Nmap is useful.
 - Perform a simple scan of your own computer.
 - Understand the importance of using Nmap responsibly.
-

Goal

Learn what Nmap is, why professionals use it, and how to perform your first safe scan.

Why This Matters

Before you can protect a network, you first need to understand what's connected to it.

Nmap helps answer questions such as:

- What devices are on this network?
- Is a computer responding?
- Which services appear to be available?

Nmap doesn't magically find security problems.

It gathers information that helps you understand a network.

Plain-English Explanation

Imagine walking through a neighborhood.

You aren't looking inside houses.

You're simply observing:

- Which houses exist.
- Which lights are on.
- Which doors appear to be open.

Nmap works in a similar way.

It gathers information about systems that respond on a network.

New Terms

- Nmap
 - Host
 - Scan
 - Target
 - Open Port
-

2.6.3 What Is Nmap?

Nmap stands for:

Network Mapper

It is one of the world's most widely used network discovery tools.

Professionals use Nmap to:

- Discover devices.
 - Learn about network services.
 - Troubleshoot networks.
 - Inventory systems.
 - Verify configurations.
 - Perform authorized security assessments.
-

2.6.3 What Nmap Does

Nmap sends carefully constructed network requests.

It then analyzes the responses.

From those responses, it can determine useful information such as:

- Whether a device is online.
- Which ports appear to be open.
- Which services may be running.

Remember:

Nmap reports what it observes.

It does **not** decide whether something is good or bad.

That's where human understanding comes in.

2.6.3 Your First Safe Scan

Today we'll scan **your own computer**.

Run:

```
nmap localhost
```



or

```
nmap 127.0.0.1
```



Both commands tell Nmap to examine your own machine.

Depending on your system, you may see:

- Open ports
- Closed ports
- Service names

Don't worry if your output looks different from someone else's.

Different computers run different services.

2.6.3 Reading the Results

You might see something like:

PORT	STATE	SERVICE
22/tcp	open	ssh

This tells us:

- Port 22 responded.
- The port appears to be open.
- Nmap believes the service is SSH.

Notice what Nmap is doing.

It is reporting observations.

It isn't attacking anything.

Learning Tip

Don't try to memorize port numbers today.

Focus on understanding what Nmap is showing you.

Looking Under the Hood

Nmap works by sending network traffic and analyzing how devices respond.

Different responses reveal different information about a system.

Why Professionals Do This

Network administrators frequently use Nmap to verify that expected services are running and to identify unexpected services that may require attention.

Using Nmap Responsibly

Nmap is an extremely useful tool.

Like many professional tools, it should only be used responsibly.

Only scan:

- Systems you own.
- Systems you have explicit permission to assess.
- Authorized training environments.

Never assume permission.

When in doubt, ask first.

Try It Yourself

Run:

```
nmap localhost
```



Read the output carefully.

Ask yourself:

- Did Nmap find any open ports?
- Which services were identified?
- Were you surprised by the results?

Don't worry if you don't recognize every service.

We'll learn about many of them throughout your cybersecurity journey.

Reflection Questions

Without looking back:

1. What does Nmap stand for?
 2. What is Nmap designed to do?
 3. What is an open port?
 4. Why should you only scan systems you own or are authorized to assess?
 5. What surprised you about your first scan?
-



Memory Helper

Nmap

→ Network discovery

Host

→ Device

Scan

→ Gather information

Open Port

→ Service accepting connections



Common Beginner Mistake

Beginners sometimes believe Nmap is a hacking tool.

Nmap is primarily an information gathering and network discovery tool.

How someone chooses to use information is what matters.

Real-World Example

Imagine walking through a parking lot.

You notice which cars are present.

You notice whether their lights are on.

You're observing—not interfering.

Nmap works in a similar way by observing network responses.

Progress, Not Perfection

You are not expected to understand every line of Nmap's output today.

The important thing is recognizing that Nmap helps you understand a network through careful observation.

Looking Ahead

Now that you've met Nmap, you'll meet another essential cybersecurity tool.

Instead of discovering devices, the next tool lets you observe network communication itself.

That tool is **Wireshark**.

Stop and Check

You are ready to continue when:

- ☐ You know what Nmap is.
- ☐ You understand what Nmap is designed to do.
- ☐ You can perform a basic scan of your own computer.
- ☐ You understand why permission matters.
- ☐ You recognize that Nmap reports observations rather than making security decisions.

Guide Summary

In this guide you learned:

- Nmap stands for Network Mapper.
 - Nmap discovers and observes network information.
 - Nmap can identify responding hosts and open ports.
 - A simple localhost scan is a safe way to begin learning.
 - Nmap should only be used on systems you own or are authorized to assess.
-

What To Do Next

Continue to **Meeting Wireshark**.

2.6.4 Guide 24: Meeting Wireshark

Estimated Time: 35–45 minutes

Difficulty: Beginner

Have You Ever Wondered...

...what computers are actually saying to each other when they communicate?

Every time you browse a website, stream a video, or send a message, your computer exchanges thousands of tiny pieces of information across the network.

Wireshark lets you observe that communication.

Learning Objectives

By the end of this guide, you should be able to:

- Explain what Wireshark is.
 - Understand what a packet is.
 - Explain why professionals use Wireshark.
 - Capture your first network traffic.
 - Recognize that encrypted traffic is still visible, even if its contents cannot be read.
-

Goal

Learn what Wireshark is designed to do and how it helps professionals understand network communication.

Why This Matters

Networks are constantly exchanging information.

Sometimes that communication doesn't work as expected.

Wireshark helps professionals observe what is actually happening instead of guessing.

Understanding network traffic is an essential cybersecurity skill.

Plain-English Explanation

Imagine standing beside a busy highway.

Cars pass by constantly.

You aren't stopping traffic.

You aren't changing anything.

You're simply watching.

Wireshark works in much the same way.

It observes network traffic as it passes through a network connection.

New Terms

- Packet
 - Capture
 - Interface
 - Traffic
 - Protocol
-

2.6.4 What Is Wireshark?

Wireshark is one of the world's most widely used network protocol analyzers.

Its primary purpose is to capture and display network traffic.

Instead of showing files or applications, it shows individual packets traveling across the network.

2.6.4 What Is a Packet?

Earlier you learned that information travels across networks.

That information is broken into many smaller pieces.

Those small pieces are called packets.

Think of shipping a large order.

Instead of putting everything into one enormous box, it is divided into many smaller packages.

Networking works similarly.

2.6.4 What Wireshark Shows You

Wireshark allows you to observe information such as:

- Source IP address
- Destination IP address
- Protocol being used
- Time the packet was seen
- Packet size

At this stage, you don't need to understand every column.

Simply recognize that Wireshark is displaying network activity.

2.6.4 Your First Capture

Open Wireshark.

You'll see a list of available network interfaces.

Choose the interface that is currently connected to your network.

For many learners, this may be:

- Wi-Fi
- Ethernet

Start a capture.

Now open a website in your browser.

Return to Wireshark.

You'll see packets appearing almost immediately.

Stop the capture after a few seconds.

Congratulations—you've captured your first network traffic.

What You'll See

Your capture will almost certainly look different from someone else's.

That's expected.

Every computer communicates differently depending on:

- Which applications are running.
- Which websites are open.
- Which devices are on the network.

The important thing is recognizing that you're observing real network communication.

2.6.4 Can Wireshark Read Everything?

No.

Many websites use encryption.

Wireshark can still show that communication is taking place.

However, it usually cannot display the protected contents of properly encrypted traffic.

That's one of the reasons encryption is so important.



Learning Tip

Don't try to understand every packet.

Start by recognizing patterns.

Understanding comes with practice.

Looking Under the Hood

Wireshark captures packets directly from your network interface.

It doesn't generate the traffic.

It simply records what your computer is already sending and receiving.

Why Professionals Do This

Network administrators and cybersecurity professionals use Wireshark to troubleshoot network problems, investigate unusual behavior, and better understand how applications communicate.

Try It Yourself

Open Wireshark.

Start a capture.

Visit one or two websites.

Stop the capture after about 30 seconds.

Look through the packets.

Can you identify:

- Source addresses?
- Destination addresses?
- Different protocols?
- Repeated communication?

Don't worry if you don't understand every detail.

Observation is the goal.

Reflection Questions

Without looking back:

1. What is Wireshark designed to do?

2. What is a packet?
3. Does Wireshark create network traffic?
4. Why might two people see different packet captures?
5. Why can't Wireshark automatically read encrypted traffic?

✓ Memory Helper

Wireshark

→ Observe network traffic

Packet

→ Small piece of network communication

Capture

→ Record network traffic

⚠ Common Beginner Mistake

Beginners often think seeing lots of packets means something is wrong.

In reality, modern computers constantly communicate across the network.

Seeing many packets is completely normal.

🔧 Real-World Example

Imagine sitting beside a highway counting vehicles.

You aren't controlling traffic.

You're simply observing what passes by.

Wireshark performs a similar role for network communication.

Progress, Not Perfection

Even experienced professionals don't understand every packet immediately.

Focus on becoming comfortable observing network traffic before learning advanced analysis techniques.

Looking Ahead

You've now met two of the most important tools in cybersecurity.

Next, you'll learn how to safely explore your own network while respecting ethics, privacy, and authorization.

Stop and Check

You are ready to continue when:

- ☐ You know what Wireshark is.
 - ☐ You understand what a packet is.
 - ☐ You can perform a basic packet capture.
 - ☐ You understand why packet captures differ from computer to computer.
 - ☐ You recognize that encrypted traffic is still visible even if its contents cannot be read.
-

Guide Summary

In this guide you learned:

- Wireshark captures and displays network traffic.
 - Network communication is divided into packets.
 - Packet captures vary from one computer to another.
 - Encryption protects the contents of much network traffic.
 - Observation is the first step toward understanding.
-

What To Do Next

Continue to **Safe Network Discovery**.

2.6.5 Guide 25: Safe Network Discovery

Estimated Time: 35–45 minutes

Difficulty: Beginner

Have You Ever Wondered...

...how cybersecurity professionals know where the line is between learning and unauthorized activity?

Many tools used by defenders can also be misused.

The difference is not the tool.

The difference is **permission**.

In this guide, you'll learn how to practice safely, ethically, and responsibly.

Learning Objectives

By the end of this guide, you should be able to:

- Explain why authorization matters.
 - Recognize safe environments for learning.
 - Understand the importance of ethical behavior.
 - Know when it is appropriate to use cybersecurity tools.
-

Goal

Develop good habits that will guide your cybersecurity journey.

Why This Matters

Cybersecurity is built on trust.

Organizations trust professionals to help protect their systems.

That trust depends on using knowledge responsibly.

Building ethical habits now will serve you throughout your career.

Plain-English Explanation

Imagine owning a flashlight.

Using it to inspect your own attic is perfectly reasonable.

Using it to look through someone else's windows without permission is not.

The flashlight didn't change.

The situation did.

Cybersecurity tools are no different.

New Terms

- Authorization
 - Permission
 - Scope
 - Assessment
 - Responsible Use
-

2.6.5 Permission Comes First

Before using any cybersecurity tool, ask yourself:

- Do I own this system?
- Do I have explicit permission?
- Am I working in an approved training environment?

If the answer is "no" or "I'm not sure," stop and ask first.

2.6.5 Safe Places to Learn

Excellent environments include:

- Your own computer
- Your own home network
- Virtual machines
- Practice labs
- Capture-the-Flag environments
- Training platforms that invite testing

These environments are designed for learning.

2.6.5 Places You Should Not Test

Do **not** assume permission simply because a system is publicly accessible.

Examples include:

- Someone else's home network
- School networks
- Workplace systems (unless authorized)
- Public Wi-Fi networks
- Websites you don't own
- Internet-connected devices that aren't yours

Visibility is **not** permission.

2.6.5 Responsible Curiosity

Curiosity is one of the most valuable qualities in cybersecurity.

Ask questions.

Experiment.

Learn.

But always do so in environments where your actions are expected and authorized.

Responsible curiosity builds trust.

Learning Tip

Whenever you're unsure whether something is appropriate to test, pause and ask.

Seeking clarification is a sign of professionalism, not weakness.

Looking Under the Hood

Many organizations clearly define what systems may be tested, what techniques are allowed, and when testing may occur.

This is often called the **scope** of an assessment.

Why Professionals Do This

Professional security assessments begin with written authorization.

This protects both the organization and the security professionals performing the work.

Try It Yourself

Think about your current environment.

Create two lists.

Safe to Practice

Examples might include:

- Your laptop
- Your desktop
- Your virtual machine
- Your home lab
- Your personal router

Not Safe to Practice

Examples might include:

- Your employer's network
- School systems
- Public Wi-Fi
- A friend's computer (without permission)

If you're ever unsure, ask before testing.

Reflection Questions

Without looking back:

1. Why is permission important?
 2. What does authorization mean?
 3. Can a publicly accessible website be tested without permission?
 4. What is a safe practice environment?
 5. Why is responsible behavior important in cybersecurity?
-



Memory Helper

Own it

or

Have permission.

If neither is true...

Don't test it.



Common Beginner Mistake

Some beginners assume that if a system responds to network traffic, it's acceptable to scan.

It isn't.

Authorization—not technical possibility—determines whether testing is appropriate.



Real-World Example

Imagine borrowing a friend's car.

Even if you know how to drive it, you still ask before taking it for a drive.

Cybersecurity follows the same principle.

Ability never replaces permission.



Progress, Not Perfection

Developing ethical habits now will help you become a trusted cybersecurity professional later.

Looking Ahead

You've learned about two powerful cybersecurity tools and how to use them responsibly.

Next, you'll complete your first home network inventory by documenting the devices you own and understand.

Stop and Check

You are ready to continue when:

- ☐ You understand why authorization matters.
- ☐ You know where it is appropriate to practice.
- ☐ You recognize the importance of responsible behavior.
- ☐ You understand that permission always comes before testing.

Guide Summary

In this guide you learned:

- Cybersecurity requires trust.
- Authorization is essential.
- Safe practice environments help you learn responsibly.
- Ethical behavior is a professional skill.

- Permission always comes before testing.
-

What To Do Next

Continue to **Your First Home Network Inventory**.

2.6.6 Guide 26: Your First Home Network Inventory

Estimated Time: 45–60 minutes

Difficulty: Beginner

Have You Ever Wondered...

...how cybersecurity professionals keep track of all the devices on a network?

Before anyone can protect a network, they first need to understand what is connected to it.

That's why creating an inventory is one of the most valuable first steps.

Learning Objectives

By the end of this guide, you should be able to:

- Explain what a network inventory is.
 - Identify devices on your home network.
 - Document basic information about those devices.
 - Understand why inventories are useful.
 - Begin thinking like a cybersecurity professional.
-

Goal

Create your first network inventory using observation and documentation.

Why This Matters

You can't protect what you don't know exists.

One of the first questions professionals ask is:

■ "What devices are actually on this network?"

Building an inventory helps answer that question.

It also gives you a useful reference if something changes later.

Plain-English Explanation

Imagine walking into a workshop.

Before organizing tools or making improvements, you'd probably take a look around and see what equipment is already there.

A network inventory is the same idea.

You're simply documenting what exists.

New Terms

- Inventory
 - Asset
 - Device
 - Host
 - Documentation
-

2.6.6 What Is a Network Inventory?

A network inventory is simply a list of the devices connected to a network.

Examples include:

- Desktop computers
- Laptops
- Phones
- Tablets
- Printers
- Smart TVs
- Streaming devices
- Game consoles

- Security cameras
 - Smart speakers
 - Network storage
 - Routers
-

2.6.6 What Should You Record?

For now, keep it simple.

Create a table like this.

Device	Connection	Approximate Location	Notes
Laptop	Wi-Fi	Office	Personal computer
Printer	Wi-Fi	Office	Inkjet printer
Smart TV	Wi-Fi	Living Room	Streaming apps
Desktop	Ethernet	Office	Main workstation

You don't need to record IP addresses yet.

We'll build on this inventory later in your cybersecurity journey.

2.6.6 Observing Your Network

Walk through your home.

Think about every device that communicates over your network.

You may be surprised by what you discover.

Examples include:

- Phones
- Tablets
- Watches
- Smart light bulbs

- Voice assistants
- Streaming sticks
- Smart thermostats
- Security systems
- Garage door openers
- Refrigerators
- Robot vacuums

Modern homes often contain dozens of connected devices.

2.6.6 Why Inventories Matter

Imagine your Wi-Fi suddenly slows down.

If you already know which devices belong on your network, unusual activity becomes much easier to notice.

Inventories also help when:

- Replacing equipment.
- Troubleshooting problems.
- Planning upgrades.
- Improving security.
- Remembering what's connected months later.



Learning Tip

Good documentation saves time.

Professionals often spend just as much time documenting systems as they do configuring them.



Looking Under the Hood

Larger organizations often maintain detailed asset inventories containing device names, IP addresses, operating systems, owners, locations, and maintenance history.

You're creating a simplified version of that same idea.



Why Professionals Do This

Asset inventories help organizations identify unknown devices, remove outdated equipment, and better understand what they're responsible for protecting.

Try It Yourself

Create your own inventory.

Include every network-connected device you can find.

For each device, record:

- Device name
- Wi-Fi or Ethernet
- Approximate location
- Short description

Don't worry if you miss one.

You can always update your inventory later.

Reflection Questions

Without looking back:

1. What is a network inventory?
2. Why is documentation important?
3. Why do professionals maintain inventories?
4. What surprised you about your own network?
5. What devices did you forget the first time you looked?

✓ Memory Helper

Inventory

→ Know what you have

Documentation

→ Keep useful records

Observation

→ Before action

⚠ Common Beginner Mistake

Many people assume they know every device on their network.

Once they begin documenting, they often discover devices they had forgotten about.

🔧 Real-World Example

Imagine buying a toolbox with fifty tools.

Before organizing it, you'd probably count the tools and see what you have.

Cybersecurity professionals approach networks in much the same way.

🔧 Progress, Not Perfection

Your inventory doesn't need to be perfect.

The important thing is building the habit of observing and documenting your environment.

Looking Ahead

Congratulations!

You've completed every technical guide in the Private Preview learning path.

Next, you'll review everything you've learned before celebrating your final milestone.

Stop and Check

You are ready to continue when:

- ☐ You understand what a network inventory is.
 - ☐ You can identify devices on your own network.
 - ☐ You understand why documentation is valuable.
 - ☐ You have created a basic inventory.
-

Guide Summary

In this guide you learned:

- What a network inventory is.
 - Why professionals document systems.
 - How to identify devices on your network.
 - Why observation and documentation are essential cybersecurity skills.
-

What To Do Next

Continue to **Module 5 Review**.

2.6.7 Module 5 Review

Congratulations!

You've completed **Your First Security Tools**.

This module introduced two of the most widely used tools in cybersecurity while reinforcing the importance of ethics, observation, and responsible decision-making.

Take a few minutes to review what you've learned before celebrating your final milestone.

2.6.7 What You Learned

During this module you learned:

- Why Kali Linux includes so many tools.
- What Nmap is designed to do.
- What Wireshark is designed to do.
- Why permission always comes before testing.
- How to begin documenting your own network.

Most importantly...

You learned that cybersecurity tools help us **observe and understand** systems—not magically solve problems.

2.6.7 Quick Review

Without looking back...

Kali

- Why does Kali Linux include so many tools?
 - Do professionals use every tool?
-

Nmap

- What does Nmap do?

- What information can it gather?
 - Why should you only scan systems you own or are authorized to assess?
-

Wireshark

- What is a packet?
 - What does Wireshark capture?
 - Does Wireshark create network traffic?
-

Ethics

- Why is authorization important?
 - What makes a safe learning environment?
-

Documentation

- What is a network inventory?
 - Why do professionals document systems?
-

2.6.7 Practice Challenge

Think about your own home network.

Can you explain:

- Which devices are connected?
- Which use Wi-Fi?
- Which use Ethernet?
- Which services you observed with Nmap?
- Why documentation makes future troubleshooting easier?

Don't worry if you don't know every answer.

Learning is a continuous process.

✓ Memory Helper

Kali

→ Cybersecurity tools

Nmap

→ Network discovery

Wireshark

→ Network observation

Permission

→ Always required

Inventory

→ Know what you have

🔥 Learning Tip

Great cybersecurity professionals ask questions before running tools.

Understanding always comes first.

📝 Progress, Not Perfection

You are not expected to become an expert after five modules.

You've built a strong foundation that will support everything you learn next.

2.6.7 Self Check

How confident do you feel?

- ☐ I'm still building confidence.
- ☐ I understand the basics.
- ☐ I can explain what these tools do.

 I'm excited to continue learning.

There are no right or wrong answers.

2.6.7 Looking Ahead

This is only the beginning.

Cybersecurity is a lifelong learning journey.

Every concept you learn will make future topics easier to understand.

What To Do Next

Continue to **Module 5 Milestone**.

2.6.8 🎉 Congratulations!

You've completed the **Cybersecurity First Steps** Private Preview learning path.

Take a moment to appreciate what you've accomplished.

When you began this journey, many of these concepts were probably unfamiliar.

Now you understand the foundations of Linux, networking, and several of the most important cybersecurity tools used by professionals around the world.

2.6.8 You Can Now...

- ✅ Start and use Kali Linux.
- ✅ Navigate the Linux terminal.
- ✅ Manage files and directories.
- ✅ Understand Linux permissions.
- ✅ Explain how networks work.
- ✅ Understand IP addresses, ports, and DNS.
- ✅ Explain the purpose of Nmap.
- ✅ Explain the purpose of Wireshark.
- ✅ Practice responsibly and ethically.
- ✅ Build a basic network inventory.

These are meaningful milestones that many beginners never take the time to learn properly.

2.6.8 Your Journey So Far

You have completed:

- ✅ Module 1 – Introduction to Cybersecurity
- ✅ Module 2 – Getting Started with Kali Linux
- ✅ Module 3 – Learning Your Way Around Linux

✓ Module 4 – Understanding Networks

✓ Module 5 – Your First Security Tools

That's an incredible accomplishment.

✓ Celebrate Your Progress

Every cybersecurity professional started as a beginner.

The knowledge you've built here forms the foundation for everything that comes next.

🔗 Progress, Not Perfection

You don't need to know everything.

The most important thing is understanding the fundamentals and continuing to practice.

2.6.8 Remember

Cybersecurity is not about memorizing commands.

It is about:

- Curiosity
- Observation
- Problem solving
- Documentation
- Ethics
- Continuous learning

Those habits will serve you far beyond this course.

2.6.8 Where Do You Go From Here?

This Private Preview is only the beginning.

Future learning paths will explore topics such as:

- Linux in greater depth
- Networking fundamentals
- Nmap
- Wireshark
- Web security
- Wireless networking
- Digital forensics
- Home labs
- Defensive security
- Scripting and automation

One step at a time.

2.6.8 Thank You

Thank you for being part of the Horizon Learning journey.

Your curiosity, patience, and willingness to learn are what truly matter.

Keep asking questions.

Keep learning.

Keep building.

We'll be here to help.

2.6.8 Ready for the Next Adventure?

Take what you've learned.

Practice it.

Build on it.

And never stop being curious.

See you in the next learning path.

2.7 Resources

2.7.1 Beginner Progress Checklist

Use this checklist to track your progress through Cybersecurity First Steps.

Module 1: Getting Started

- ☐ Lesson 1: Learning Path
- ☐ Lesson 2: What Is Cybersecurity?
- ☐ Lesson 3: Safety and Permission

Module 2: Kali Linux

- ☐ Lesson 4: What Is Kali?
- ☐ Lesson 5: Download Kali
- ☐ Lesson 6: Create a Live USB
- ☐ Lesson 7: First Boot

Module 3: Linux Basics

- ☐ Lesson 8: Terminal Basics
- ☐ Lesson 9: Files and Folders
- ☐ Lesson 10: Useful Commands

Module 4: Networking Basics

- ☐ Lesson 11: Networking Basics
- ☐ Lesson 12: Wi-Fi Basics
- ☐ Lesson 13: IP Addresses and Ports

Module 5: Beginner Tools

- ☐ Lesson 14: Kali Tools Overview
- ☐ Lesson 15: Nmap
- ☐ Lesson 16: Wireshark

Module 6: Safe Practice Labs

- ☐ Lesson 17: Home Wi-Fi Security Check
- ☐ Lesson 18: First Nmap Scan
- ☐ Lesson 19: Beginner Lab Roadmap

Beginner skills checkpoint

By the end of this learning path, you should be able to:

- ☐ Explain what cybersecurity means
- ☐ Explain why permission matters
- ☐ Describe what Kali Linux is used for
- ☐ Open a Linux terminal
- ☐ Use basic commands like `pwd`, `ls`, and `cd`
- ☐ Explain what an IP address is
- ☐ Explain what a port is
- ☐ Run a basic Nmap scan against your own computer or router
- ☐ Open Wireshark and recognize that it shows network traffic
- ☐ Complete a safe home Wi-Fi security review

2.7.2 Glossary

This glossary explains beginner terms in plain English.

Cybersecurity

The practice of protecting computers, networks, accounts, and data.

Kali Linux

A Linux system made for cybersecurity learning, testing, and security work.

Linux

An operating system, like Windows or macOS, but commonly used for servers, development, and security work.

Terminal

A text-based way to control the computer by typing commands.

Command

An instruction typed into the terminal.

Directory

Another word for folder.

File path

The location of a file or folder on the computer.

Root

The highest-level administrator account on Linux.

Sudo

A command that lets an approved user run something with administrator-level permissions.

ISO

A disk image file. In this guide, the Kali ISO is used to create a live USB.

Live USB

A USB drive that can start an operating system without installing it permanently.

Router

The device that connects your home network to the internet.

IP address

A number that identifies a device on a network.

Port

A numbered doorway that a device or program uses for network communication.

Service

A program running on a device that listens for network connections.

Scan

A check performed by a tool to learn information about a system or network.

Vulnerability

A weakness that could cause a security problem.

Exploit

A method used to take advantage of a vulnerability.

3 About Horizon Learning

Horizon Learning exists to make complex technology approachable.

Many people want to learn topics like Linux, networking, cybersecurity, programming, and self-hosting, but the learning process can feel scattered and intimidating.

Horizon Learning is designed to provide a smoother path.

3.1 Mission

Make practical technology education easier to understand, easier to follow, and easier to apply.

3.2 Who this is for

Horizon Learning is for people who:

- Are new to technical topics
- Prefer plain-English explanations
- Want step-by-step guidance
- Want to understand what they are doing, not just copy commands
- Want safe hands-on practice
- Want a clear path from beginner to confident learner

3.3 What makes this different

Horizon Learning is not trying to be the biggest library of tutorials on the internet.

The goal is to be organized, practical, and beginner-friendly.

Each learning path is designed to answer:

- What is this?
- Why does it matter?
- How does it work?
- How do I practice it safely?
- What should I learn next?

3.4 Learning approach

Horizon Learning focuses on:

- Fundamentals before advanced topics
- Understanding before memorization
- Practical examples
- Short lessons
- Clear checkpoints
- Safe practice environments
- Responsible technology use

3.5 Current focus

The first learning path is:

- Cybersecurity First Steps

This path introduces cybersecurity, Kali Linux, Linux basics, networking fundamentals, beginner tools, and safe practice labs.

3.6 Long-term direction

Horizon Learning may eventually include more learning paths, downloadable guides, checklists, glossaries, labs, and interactive progress tracking.

For now, the priority is simple:

Build one clear, useful, beginner-friendly learning path at a time.

4 Teaching Philosophy

Horizon Learning is built around one idea:

Technical subjects become easier when they are explained clearly, in the right order, with practical examples.

4.1 Explain, do not impress

Guides should help the learner understand.

The goal is not to sound advanced.

The goal is to make the topic usable.

Avoid unnecessary jargon. When technical terms are needed, explain them.

4.2 Never assume prior knowledge

A beginner may not know what Linux is, what a terminal is, what an IP address is, or why a command matters.

That is okay.

Guides should meet the learner where they are.

4.3 Teach the why

Do not only say what to do.

Explain why the step matters.

Instead of only writing:

```
Run ip addr
```



Explain:

```
This command shows network address information for the computer. It helps you find the IP address  
used on your local network.
```

4.4 Build from fundamentals

Advanced topics should come after the basics.

A learner should not be asked to use a tool before they understand the concept behind it.

For example:

- Learn what an IP address is before scanning a network.
- Learn what a port is before reading Nmap output.
- Learn what a packet is before using Wireshark.
- Learn basic Linux commands before using Kali tools.

4.5 Learn by doing

Reading is useful, but practice builds confidence.

Guides should include safe hands-on activities when possible.

Good practice activities are:

- Small
- Clear
- Authorized
- Reversible
- Easy to check

4.6 Safety first

Some technical tools can affect systems, networks, or data.

Guides should clearly explain safe boundaries.

For cybersecurity topics, the rule is simple:

Only test systems, networks, accounts, and devices that you own or have clear permission to test.

4.7 Practical over theoretical

Theory matters, but beginners need practical connection.

Explain concepts in a way that connects to real tasks.

Example:

A port is not just a number. It is a way for a device to receive a specific type of network communication.

4.8 Confidence matters

Learners often quit because they cannot tell if they are making progress.

Every guide should include checkpoints so the learner knows when they are ready to continue.

4.9 Clear next steps

A good guide should not leave the learner wondering what to do next.

Each guide should point naturally to the next guide, lab, checklist, or resource.

5 Horizon Learning Framework

5.1 Why Horizon Learning Exists

Technology has never been more accessible.

Learning technology has never been more fragmented.

Information is everywhere.

Understanding is not.

Horizon Learning exists to organize technology education into clear, practical, beginner-friendly learning paths that help people build real understanding and confidence.

Our goal is not to create the largest library of lessons.

Our goal is to create learning experiences that people trust.

5.2 Mission

Help people understand technology through clear explanations, practical experience, and responsible learning.

5.3 Vision

Create a technology learning platform that grows with technology while remaining approachable for beginners.

Every lesson should make learners feel more confident than they did before they started.

5.4 Core Principles

1. Understanding Before Memorization

Learners should understand why something works before trying to memorize commands or definitions.

Understanding creates confidence.

Memorization follows naturally.

2. Fundamentals Before Complexity

Advanced topics are built on simple ideas.

Never skip foundational concepts.

Never assume prior knowledge.

3. Learn by Doing

Reading is only part of learning.

Every learning path should include meaningful hands-on practice.

Small successes build confidence.

4. Learn Safely

Technology should be explored responsibly.

Cybersecurity lessons must reinforce legal and ethical boundaries.

Only practice on systems you own or have explicit permission to test.

5. Explain in Plain English

Complex topics should become easier, not more intimidating.

Use technical terminology when appropriate, but explain it clearly.

6. Build Confidence

Learners should finish each lesson feeling more capable than when they started.

Progress matters.

Confidence matters.

7. Respect the Learner

Never shame beginners.

Never assume a question is obvious.

Curiosity should always be encouraged.

8. Stay Current

Technology changes.

Education should evolve with it.

Horizon Learning is a living curriculum that will continue to improve over time.

5.5 Quality Standard

Before a lesson is considered complete, it should answer:

- What is this?
- Why does it matter?
- How does it work?
- How can I practice it safely?
- What should I learn next?

5.6 Writing Standard

Every lesson should strive to include:

- Goal
- Estimated time
- Difficulty
- Learning objectives
- Plain-English explanation
- New terms
- Step-by-step guidance
- Hands-on practice
- Reflection
- Common mistakes
- Troubleshooting
- Summary
- Checkpoint
- Next lesson

5.7 Long-Term Commitment

Horizon Learning is designed to be:

- Beginner friendly
- Practical
- Mobile friendly
- Printable

- Downloadable
- Accessible
- Technically accurate
- Continuously improving

5.8 Success Is...

Success is not measured by the number of lessons.

Success is measured by how many people finish a lesson believing:

I finally understand this.

6 Downloads

Downloadable versions of guides will be listed here.

6.1 Cybersecurity First Steps

PDF version coming soon.

Planned download:

- Cybersecurity First Steps PDF
- Beginner progress checklist
- Home Wi-Fi security checklist

6.2 Why downloads matter

Some readers may want to read the guide on a phone or tablet before trying the hands-on steps.

The online version will stay updated, while PDF versions can be used offline.

7 Updates

This page will list important updates to Horizon Learning guides.

7.1 Current status

Horizon Learning is in early setup.

The first guide being created is Cybersecurity First Steps.

7.2 Planned updates

- Add screenshots
- Add downloadable PDF guides
- Add beginner checklists
- Add more Kali setup detail
- Add beginner labs
- Add glossary terms
- Add mobile reading improvements

8 Changelog

8.1 2026-06-30

- Created Horizon Learning project.
- Added Cybersecurity First Steps structure.
- Added beginner safety page.
- Added learning path.
- Added Kali introduction.
- Added download and live USB starter pages.
- Added beginner progress checklist.

9 Content Roadmap

This page tracks what still needs to be written.

9.1 Version 0.1

Goal: create a complete beginner-readable guide with placeholder content on every page.

- ☐ Homepage
- ☐ Cybersecurity First Steps overview
- ☐ Safety and Permission
- ☐ Learning Path
- ☐ What Is Cybersecurity?
- ☐ What Is Kali?
- ☐ Download Kali
- ☐ Create Live USB
- ☐ First Boot
- ☐ Terminal Basics
- ☐ Files and Folders
- ☐ Useful Commands
- ☐ Networking Basics
- ☐ Wi-Fi Basics
- ☐ IP Addresses and Ports
- ☐ Kali Tools Overview
- ☐ Nmap
- ☐ Wireshark
- ☐ Home Wi-Fi Security Check
- ☐ First Nmap Scan
- ☐ Beginner Lab Roadmap
- ☐ Downloads page
- ☐ Changelog page

9.2 Version 0.2

Goal: improve the guide after beginner feedback.

- ☐ Add screenshots
- ☐ Add PDF download
- ☐ Add printable checklists
- ☐ Add glossary expansion
- ☐ Add beginner quizzes
- ☐ Add clearer troubleshooting sections

9.3 Future portal ideas

- User accounts
- Saved progress
- Completed lessons
- Notes
- Bookmarks
- Interactive checklists
- Downloadable files inside the portal

10 Horizon Learning Style Guide

This style guide defines how Horizon Learning lessons should be written.

The goal is consistency, clarity, and beginner-friendly education.

10.1 Standard lesson structure

Each lesson should use this structure when possible:

1. Goal
2. Why this matters
3. Plain-English explanation
4. New terms
5. Prerequisites
6. Step-by-step walkthrough
7. Try it yourself
8. Common mistakes
9. Troubleshooting
10. Stop and check
11. Summary
12. What to do next

Not every lesson needs every section, but most lessons should follow this pattern.

10.2 Writing rules

Assume the learner is new

Do not assume the reader knows Linux, networking, programming, cybersecurity, Git, Docker, or command-line tools.

If a term is important, explain it.

Keep paragraphs short

Long blocks of text are harder to read, especially on mobile.

Prefer short paragraphs.

Avoid unnecessary jargon

Use technical terms when needed, but explain them.

Bad:

```
Enumerate exposed services on the target host.
```



Better:

```
List the services that are available on the device you are allowed to test.
```



Explain commands before or after showing them

Do not show commands without context.

Bad:

```
ip addr
```



Better:

```
Run this command to show network address information:
```



```
ip addr
```

Avoid “simply”

Do not say:

```
Simply run this command.
```



A step may not feel simple to a beginner.

Instead say:

```
Run this command.
```



Then explain what it does.

Avoid shame

Do not write in a way that makes beginners feel bad for not knowing something.

Avoid phrases like:

- Obviously
- Everyone knows

- It should be easy
- Just do this
- Simple

Use plain-English examples

Whenever possible, compare technical ideas to familiar ideas.

Example:

An IP address is like a device address on a network.

A port is like a specific door at that address.

Use safe boundaries

Security lessons must clearly explain what is allowed.

For cybersecurity content, include permission reminders before hands-on labs.

Prefer small steps

Break large tasks into smaller steps.

A beginner should be able to follow the page without guessing.

10.3 Checkpoint style

Use checklists for checkpoints.

Example:

You are ready to continue when:

- ☐ You understand what the command does.
- ☐ You ran it successfully.
- ☐ You know what result to expect.
- ☐ You know what to do next.

10.4 Lab style

Labs should include:

- Goal

- Permission requirement
- What you need
- Steps
- Expected result
- How to record findings
- Common mistakes
- Stop and check

10.5 Safety language for cybersecurity

Use this wording when needed:

Only test systems, networks, accounts, and devices that you own or have clear permission to test.

If you are unsure whether you have permission, stop.

10.6 Mobile-friendly writing

Many learners may read lessons on a phone before practicing on a computer.

To support mobile readers:

- Keep sections short
- Use clear headings
- Use bullet lists when helpful
- Avoid very wide tables
- Avoid long code blocks when possible
- Keep steps numbered and direct

10.7 Tone

The tone should be:

- Clear
- Calm
- Practical
- Encouraging
- Direct
- Beginner-friendly

The tone should not be:

- Condescending
- Overly academic
- Overly casual
- Fear-based
- Sensationalized